# Improving Lightweight Super-Resolution via Depth Compression: A Statistically-Validated Accuracy–Efficiency Trade-off for Ear Biometrics


**Abstract**

Ear recognition is gaining attention as a complement to face recognition, but ear images are often captured well below the resolution lightweight super-resolution (SR) architectures such as SPAN were designed for. This paper asks how far such a model can be compressed before it stops helping, and how much of the resulting gap SR closes, answering both with **span_tiny**, a depth-compressed SPAN variant (3 attention blocks instead of 6, a 46.0% parameter reduction) evaluated across eleven backbones with multi-seed testing. On the primary dataset (EarVN1.0), span_tiny and the uncompressed baseline both significantly improve identity accuracy over no-SR in nearly every backbone; the accuracy cost of compression reaches significance in only one of eleven backbones and is negligible overall (equivalence test, $p = 0.008$). A frequency-domain test explains why: removing the lowest 10% of the spectrum collapses accuracy to near chance in all eleven backbones, while keeping only that band preserves a quarter of it, so identity-relevant signal survives compression because it lives in coarse, low-frequency structure. Against a high-resolution ceiling, SR recovers only a fifth to a quarter of the gap, matched almost exactly by the uncompressed baseline. The compression-safety finding replicates on a second dataset, AWEx (336 subjects, three protocols, zero of eleven backbones showing a cost), and against five other lightweight SR architectures span_tiny is never beaten, tying the top performers at a fraction of their parameter count. One exception: on low-resolution images, one backbone (ResNet-18) reverses this ordering, traced to a train/test resolution mismatch.



**Keywords:** lightweight super-resolution, depth compression, ear recognition, low-resolution biometrics, equivalence testing


# 1 Introduction

## 1.1 Background

Ear biometrics has drawn growing interest as a complement or alternative to face recognition, particularly since widespread mask-wearing made face-based identification unreliable in many settings [1]. The ear offers several properties useful for identification: its structure remains largely stable across adulthood, it carries a distinctive pattern of cartilage folds, and it changes little with facial expression, unlike the face. However, ear recognition in unconstrained settings (surveillance footage, images captured at a distance) routinely produces ear regions at very low pixel resolution, often well below what recognition backbones or SR networks were designed for.

Deep-learning-based SR began with Dong et al.'s SRCNN [2], a shallow end-to-end CNN mapping directly from LR to HR pixels, and was made computationally practical at higher resolutions by Shi et al.'s sub-pixel convolution [3], which moved upsampling to the final layer via a learned pixel-shuffle operation, the same upsampling mechanism SPAN itself uses (Section 3.2). A separate branch pursued perceptual realism over pixel fidelity: Ledig et al.'s SRGAN [4] combined an adversarial loss with a perceptual feature loss to produce photo-realistic detail at high upscaling factors, an objective distinct from the pixel- and distillation-based training recipe used in this paper (Section 3.3), where the downstream task rather than perceptual quality is the target. Lightweight SR has advanced considerably since, driven largely by the NTIRE efficient super-resolution challenge series. SPAN [5] is among the most efficient recent designs. It won both the overall-performance and runtime tracks of NTIRE 2024 by replacing learned attention (of the kind popularized for vision by squeeze-and-excitation channel gating [6]) with a symmetric, parameter-free activation applied directly to convolutional outputs. Other designs pursue efficiency through different mechanisms: RLFN [7] simplifies feature aggregation, ECBSR [8] uses hardware-friendly re-parameterized convolutions in the same general spirit as RepVGG's structural re-parameterization for classification backbones [9], SAFMN [10] relies on spatially-adaptive modulation instead of self-attention, and SMFANet [11] combines self-modulation with a lightweight feed-forward design. All of these architectures are designed and benchmarked at input resolutions of roughly 32–64 px or higher, well above the $\sim 20\times 20$-pixel regime addressed in this paper, and none has been evaluated for further depth compression under statistically rigorous validation against a concrete downstream task.

## 1.2 Research gap

Two gaps motivate this work. First, most lightweight SR papers report PSNR and SSIM on standard benchmarks such as Set5, Set14, Urban100, and DIV2K. These metrics say little about whether a compressed model remains useful as a pre-processing step for a specific recognition task at extremely low input resolution. Some precedent exists for optimizing SR toward a downstream task rather than pixel fidelity: Ataer-Cansizoglu et al. [12] demonstrated this for low-resolution face verification. Almost no equivalent work exists for ear biometrics, and none addresses how far a lightweight architecture can be compressed before downstream accuracy degrades. Second, papers in this area typically report a single training run without a variance estimate, making it difficult to determine whether a reported accuracy gap reflects a genuine effect or seed-to-seed noise. This concern is more acute for downstream classification accuracy than for SR pixel metrics, since classification accuracy is a noisier signal, and a single run therefore provides weaker support for a claim than it would for PSNR.

### 1.3 Contributions

This paper's central contribution is **span_tiny**, a depth-compressed variant of SPAN (`n_blocks` reduced from 6 to 3, `feat`=48) that cuts parameter count by 46.0% (0.230M vs. 0.4263M) and measured inference latency by 54.8% (0.256 ms vs. 0.566 ms, single development GPU, Table 10; not a deployment benchmark, Section 7); the re-implementation is validated against the original authors' published SPAN-S results before being applied to the new task, matching 99.93% of the reported benchmark PSNR. Four further results carry most of the paper's evidentiary weight. This paper asks a single, pre-registered question: does depth compression preserve the ordering `no-SR` < `span_tiny` < `uncompressed baseline`? Multi-seed ($n = 5$), multi-backbone (11 recognition architectures: 5 used from the outset, plus 6 added afterward to test generality across architecture families) statistical testing shows that both span_tiny and the uncompressed baseline beat no-SR significantly in all eleven backbones on the primary dataset, EarVN1.0, while the accuracy cost of compression itself reaches significance in only one. The same compression-safety finding holds on a second, independent dataset (AWEx, 336 subjects) across three separate evaluation protocols (from-scratch training, full-fine-tuning transfer from EarVN1.0, and frozen-backbone transfer), with no backbone showing a significant accuracy cost under any of the three, evidence of generalization beyond a single dataset or transfer strategy; the secondary question of whether SR helps at all over no-SR is, by contrast, less consistently powered on this smaller dataset. The strongest single piece of evidence for the central claim comes from a depth ablation that isolates block count alone (span_tiny against span_large, an otherwise identical, uncompressed-depth counterpart), which shows no statistically significant difference in ten of eleven backbones on EarVN1.0 (the exception, EfficientNet-B0, shows a small but significant cost under one percentage point) and in all eleven on AWEx. The fourth result answers not whether compression is safe but why: a direct frequency-domain ablation (Section 5.8.2) filters the input spectrum and re-evaluates each backbone's already-trained recognition model, no re-training, isolating what the model's own predictions depend on. Removing the lowest 10% of the spectrum collapses accuracy to 2.9–5.1% of baseline in all eleven backbones; retaining only that lowest 10% preserves 22.3–33.6% of it, with zero exceptions across backbones. This is direct, cross-backbone evidence that the recognition-relevant signal in a low-resolution ear crop is concentrated in coarse, low-frequency structure a shallow network already captures — a mechanistic account for the compression-safety result above, not merely a further instance of it.

Several secondary results round out the picture without carrying the same weight. An ablation of span_tiny's loss components (pixel loss, distillation, and an additional identity term, in all four pairwise combinations) finds that the identity term significantly reduces downstream accuracy (consistent with an independent hyperparameter sweep reported in Section 3.3) while the distillation term's own contribution remains statistically indistinguishable from pixel-only training. Set against five other lightweight SR architectures (RLFN, RLFN-adapted, ECBSR, SAFMN, SMFANet) trained under an identical distillation recipe, span_tiny is never significantly beaten on downstream accuracy by any of them: it is statistically tied with the two most accurate alternatives and significantly ahead of the two least accurate, despite ranking low among the six on raw PSNR/SSIM, a sign that this work is competitive on the metric the task actually cares about, not only on efficiency. Three further mechanisms designed to improve on span_tiny beyond its core block-count reduction (a feature-level distillation loss, a saliency-weighted identity-critical loss, and learned/differentiable block pruning) were each validated to the same statistical standard as the main result and found to help not at all: the first two show no detectable effect in either direction, and learned pruning performs significantly worse than span_tiny's fixed block selection in nearly every backbone at an equal parameter count. An ablation designed to test this paper's own initial candidate explanation for why depth reduction is tolerable (that SPAN's attention mechanism carries no learned parameters) finds no support for it: two architectures with learned attention or modulation, subjected to the same proportional depth cut under the same protocol, tolerate it just as well as span_tiny does, with no significant difference in any of the 22 backbone–architecture combinations tested (all eleven backbones against both architectures). This rejection is what motivated the direct frequency-domain test of the replacement account described above (the paper's fourth central result), rather than leaving that account as an untested assumption.

To support reproducibility, the complete source code, pipeline scripts, and a step-by-step reproduction guide are publicly available at https://github.com/quangthai121121/earsr_span.

### 1.4 Paper organization

Section 2 reviews related work in ear recognition, efficient super-resolution, SR for downstream recognition, and knowledge distillation. Section 3 describes the span_tiny architecture, training recipes, and statistical methodology. Section 4 details the experimental setup across both datasets. Section 5 reports the main and contextual results, a set of three additional mechanisms tested for further improvement and found negative or null (Section 5.7), and three further questions about span_tiny's specific design choices, all now resolved: a completed attention-parameterization ablation, a completed block-removal position ablation, and a resolved-by-inspection recognition-embedding-distillation question (Section 5.8). Section 6 discusses the findings, Section 7 states limitations, and Section 8 concludes.

## 2 Related Work

### 2.1 Ear recognition

Ear recognition moved from handcrafted descriptors toward deep-learning approaches over the past decade. Alshazly et al. [13] trained an ensemble of CNNs for unconstrained ear recognition and reported that data augmentation and transfer learning were essential given the small size of available ear datasets. This constraint still shapes experimental design in this area, including the dataset choices made in this paper (Section 4.1). Interest in ear biometrics grew further once widespread mask-wearing made face-based identification unreliable in many settings. Mehta and Singh [1] developed a lightweight, augmentation-based CNN for 2D ear recognition explicitly motivated by this scenario, targeting deployment on CPU-only embedded hardware. Lendering et al. [14] proposed EdgeEar, a hybrid CNN–transformer architecture with low-rank linear layers that cuts parameter count by roughly 50× relative to prior state-of-the-art ear recognition models while achieving the lowest equal-error rate on the UERC2023 benchmark [15]. This shows that the push toward efficient, edge-deployable models seen in SR (Section 2.2) has a direct counterpart on the recognition side of the pipeline studied here. Benzaoui et al. [16] provide a broader survey of 2D, 3D, and hybrid ear recognition approaches, databases, and open challenges. More recent work continues to push both accuracy and robustness on this modality: Peddi et al. [17] propose a prototype-node graph neural network for multi-impression ear recognition, reporting rank-1 accuracy above 99% on several benchmarks; Arun et al. [18] show that overlapping-patch tokenization substantially improves Vision Transformer ear verification, including on EarVN1.0; Ozturk et al. [19] demonstrate that treating left and right ears as distinct classes during training improves CNN-based ear recognition, direct evidence for the left-right asymmetry this paper also relies on when choosing not to apply horizontal-flip augmentation (Section 4.4); and Mohamed et al. [20] report near-ceiling accuracy on EarVN1.0 itself under full-resolution, unconstrained-augmentation training, a regime very different from the extreme $20 \times 20$-pixel setting studied here (Section 5).

Publicly available ear datasets vary substantially in scale, acquisition conditions, and subject count. EarVN1.0 [21] is a large-scale, in-the-wild dataset of 28,412 color images from 164 Vietnamese subjects (98 male, 66 female), collected under unconstrained illumination and pose. The Annotated Web Ears (AWE) dataset [22] and its extensions (collectively AWEx in this work) were assembled by the University of Ljubljana ear-recognition research group, the same group behind EdgeEar [14], from web-crawled images of public figures. The base AWE set contains 1,000 images from 100 subjects. This paper extends it with the CVL Ear dataset (16 subjects, 804 images) and an additional "New Images" collection (220 subjects, 2,200 images) from the same research group, giving a combined pool of 336 subjects and 4,004 images (Section 4.1).

### 2.2 Lightweight/efficient super-resolution

Recent efficient SR research has been substantially driven by the NTIRE efficient super-resolution challenge series. RLFN [7] simplifies feature aggregation to three convolutional layers per residual block, winning the runtime track of NTIRE 2022. ECBSR [8] proposes edge-oriented convolution blocks that re-parameterize multiple operations into a single 3×3 convolution at inference time, targeting real-time performance on commodity mobile SoCs. SAFMN [10] introduces a spatially-adaptive feature modulation mechanism that captures non-local dependencies without the computational cost of dot-product self-attention, achieving parameter counts roughly 3× smaller than IMDN at comparable quality. SMFANet [11] combines a self-modulation feature aggregation module with a partial-convolution feed-forward network to balance local and non-local feature interaction efficiently. SPAN [5], the architecture compressed in this work, replaces learned attention with a parameter-free mechanism based on symmetric activation functions applied to convolutional outputs, winning both the overall-performance and runtime tracks of NTIRE 2024 [23]. The challenge itself has continued past SPAN's 2024 edition, with the 2025 edition [24] reporting incremental gains from newer entrants rather than a change in the field's overall efficiency frontier. Several further lightweight designs have appeared since SPAN, exploring complementary directions: involution combined with blueprint-separable convolutions [25], hybrid channel/spatial attention under a fully separable-convolution backbone [26], Mamba-based state-space blocks compressed via low-rank knowledge distillation for edge deployment [27], and lightweight transformers trained without access to genuine high-resolution supervision [28]. All of these, like the architectures above, target standard SR benchmarks at input resolutions well above the regime studied here, and none has been evaluated for depth compression below its original design point under a downstream recognition objective at extremely low input resolution ($\approx 20 \times 20$ px).

### 2.3 Super-resolution for downstream recognition

Recognition accuracy on low-resolution imagery does not follow directly from pixel-fidelity metrics. Wang et al. [29] showed this directly for very-low-resolution recognition (face identification, digit recognition, and font recognition), finding that networks trained and evaluated with resolution-appropriate objectives outperform those optimized for a fixed higher resolution. A separate line of work optimizes SR explicitly for downstream recognition accuracy instead of pixel fidelity. Ataer-Cansizoglu et al. [12] trained an identity-preserving face SR network to minimize distance in a pre-trained face-recognition feature space instead of pixel space, and found that this recognition-aware objective outperformed conventional pixel-fidelity SR for very-low-resolution face verification. The same question has been asked for iris biometrics. Nguyen et al. [30] showed that SR tailored for recognition at a distance improves iris matching more than generic SR does. Ribeiro et al. [31] asked directly whether photo-realistic reconstruction is necessary for iris recognition, and found that the two are related but not the same objective, matching the finding Ataer-Cansizoglu et al. report for faces. The qualitative results in this paper echo the same pattern (Figure 2b, Section 5.1): a near-tied, relatively high PSNR turns out to be driven by accurate reconstruction of hair and background rather than of the ear itself. Ear biometrics has not yet been examined through this lens. The evaluation of span_tiny directly on downstream identity accuracy, rather than on PSNR/SSIM alone, extends the same question to a modality where it has not previously been asked.

### 2.4 Knowledge distillation in super-resolution

Teacher–student distillation [32] is a common strategy for training compact SR networks, typically combining a pixel-reconstruction loss with a feature- or output-level distillation loss against a larger teacher network. The idea of supervising a compact student's intermediate representations directly, rather than only its final output, traces to Romero et al.'s FitNets [33]; the feature-level distillation term evaluated as "KD v2" in this paper's own negative-result ablation (Section 5.7) is a direct descendant of this idea, applied at the SR feature level instead of a classification backbone's. Lee et al. [34] distilled intermediate decoder features from a teacher trained with privileged access to the HR image, letting the student recover high-frequency detail it could not learn from LR input alone. Zhang et al. [35] argued that a core assumption behind most SR distillation methods does not hold: because a teacher's own SR output is itself only a noisy approximation of the true HR image, naively matching student outputs to teacher outputs caps how much the student can improve. Their proposed data-upcycling scheme was designed specifically to work around this ceiling. Jiang et al. [36] instead pool supervision from multiple teacher SR networks simultaneously via a feature-alignment strategy that reconciles their differing representations, an orthogonal direction to the single-teacher recipe used in this paper; for a broader treatment of distillation mechanisms across modalities, see the recent survey by Mansourian et al. [37]. This observation offers a useful lens on the findings reported here: the loss-ablation in Section 5.3 finds no statistically significant contribution of the distillation term to downstream accuracy at this compression level, which is consistent with distillation gains for SR being real but small and hard to detect against a noisy teacher signal, rather than with distillation being useless. The recipe used in this paper (Section 3.3) is simpler than either of these, and this paper additionally reports a rigorous ablation of its contribution to *downstream recognition accuracy* specifically, a question this literature has not directly addressed.

### 2.5 Positioning

To our knowledge, no prior work combines further depth compression of a state-of-the-art parameter-free-attention SR architecture, evaluation at an extremely low input resolution (20×20 px) substantially below the design regime of the compared architectures, and multi-seed, multi-backbone statistical validation against a downstream ear-recognition task, confirmed across two independent datasets. This combination of experimental conditions is not itself the contribution: what it enables is a testable account of *why* depth compression is tolerable here at all – that the recognition-relevant signal in a low-resolution ear crop is concentrated in coarse, low-frequency structure a shallower network already captures, a candidate explanation tested directly, both against two architectures with learned (rather than parameter-free) attention and via direct frequency-domain ablation of what a trained recognition model's own predictions depend on, rather than assumed (Section 6). This paper addresses both the experimental combination and this candidate mechanistic account.

## 3 Methodology

### 3.1 Problem formulation

Given a high-resolution ear crop, a low-resolution counterpart is generated via bicubic downsampling at a fixed scale factor ×4 (HR 80×80 → LR 20×20 on EarVN1.0; HR 76×76 → LR 19×19 on AWEx, since `hr_size` is auto-selected per dataset, Section 4.1). An SR model upsamples the LR image back to the dataset's HR size, and the resulting image is passed to a recognition backbone trained to predict subject identity and, secondarily, gender. Three domains are compared throughout: `lr` (no SR, the

LR image bicubically upsampled and fed directly to the recognition backbone, establishing the no-SR baseline), `span_baseline` (the uncompressed SPAN architecture, referred to as `sr_baseline` in project logs), and `span_tiny` (the depth-compressed variant proposed here, referred to as `sr_improved`).

The SPAN-family names used throughout this paper are collected in Table 1 for reference, since the same underlying architecture is instantiated in several configurations that differ only in block count or in which of the two implementations (custom re-implementation vs. original authors' code) is used, and these are each introduced separately across this and later sections.

**Table 1** SPAN-family terminology used throughout this paper. All configurations share the same overall architecture (Figure 1); they differ only in block count (`n_blocks`) and in which of two implementations is used (Section 3.2).

| Paper name | Blocks / implementation | Role |
|---|---|---|
| `span_tiny` | 3 blocks, custom re-implementation | This paper's proposed, depth-compressed model |
| `span_baseline` | 6 blocks, original authors' code | Uncompressed reference in the main comparison ( |
| `span_large` | 6 blocks, custom re-implementation | Uncompressed counterpart isolating depth alone ( |
| `lr` | – (no SR) | Bicubic-upsampled input, no-SR baseline |

### 3.2 span_tiny architecture

span_tiny is a depth-compressed variant of SPAN [5]. Where the uncompressed baseline uses `n_blocks=6` Swift Parameter-free Attention Blocks (SPABs) at `feat=48` channels, span_tiny uses `n_blocks=3` at the same channel width, reducing deploy-mode parameters from 0.4263M to 0.230M (−46.0%). Each SPAB (Figure 1a) passes its input $x$ through three sequential 3×3 convolutions, the first two followed by GELU, to obtain raw features $h$. These raw features feed two parallel, independently-computed paths: a residual sum $u = x + h$, and a parameter-free attention map $v = \text{sigmoid}(h) - 0.5$, a fixed activation symmetric about the origin with no learned parameters. The block output is formed by element-wise multiplication applied last, output $= u \otimes v$: the residual is added before, not after, the attention modulation. Each SPAN stack additionally carries a block-level global residual, in which the feature map immediately after the initial shallow feature-extraction convolution is added back after the final feature-aggregation convolution, immediately before pixel-shuffle upsampling (Figure 1b,c). Halving the number of SPABs removes attention *computation* across the removed blocks but not attention *parameters*, since the mechanism has none. This property is proposed in Section 6 as a partial explanation for why the ablation in Section 5.2 shows minimal accuracy degradation from depth reduction.

Before applying span_tiny to the ear-recognition task, the re-implementation was validated against the original authors' published results. On the SPAN-S benchmark configuration reported in [5] (Table 1), the re-implementation reproduces 99.93% of the reported PSNR, indicating that observed differences in the ear-recognition experiments reflect the compression itself rather than an implementation discrepancy.

span_tiny and span_large (the uncompressed-depth counterpart used in the depth ablation, Section 5.2) are both instantiated from the same custom `SPAN` class shown in Figure 1a, using plain 3×3 convolutions and differing only in `n_blocks` (3 vs. 6). This makes the span_tiny-vs-span_large comparison (Table 6) a clean, single-variable isolation of depth. By contrast, `span_baseline` (used in Table 4 as the uncompressed reference and referred to as `span_official` in project logs) is instantiated from the original authors' SPAN codebase, which internally uses a reparameterizable Conv3XC convolution design, in the same structural-reparameterization spirit as RepVGG [9], rather than plain 3×3 convolutions. Both implementations are architecturally equivalent at deploy time, since Conv3XC collapses to an equivalent single convolution after re-parameterization, and both are validated against the same published SPAN-S benchmark numbers, but they are not byte-for-byte the same low-level implementation. This is the source of the small parameter-count difference between span_large (0.417M, plain-conv 6-block) and span_baseline (0.4263M, Conv3XC-based 6-block) visible in Table 10, despite both being nominally 6-block, 48-channel configurations. Table 6 (span_tiny vs. span_large) is therefore treated as the methodologically cleaner test of the depth-compression hypothesis, while Table 4 (span_tiny vs. span_baseline) answers the closely related but distinct practical question of how the compressed model compares against deploying the official pre-trained baseline. This distinction is made explicit here so that neither table is over-interpreted as isolating a variable it does not fully isolate.

### 3.3 Training recipe: Track A vs. Track B

Two training tracks are used throughout:

- **Track A** (pixel-loss only, L1 reconstruction loss against the HR ground truth): used exclusively to cross-check SR-quality numbers against values reported in the SR literature. Track A results are not used for any comparative claim between architectures.
- **Track B** (distillation recipe, used for all comparative claims): a pre-trained, well-converged `span_official` (the original authors' SPAN implementation) serves as teacher, with loss weights $\lambda_{\text{pixel}} = 1.0$, $\lambda_{\text{distill}} = 1.0$, $\lambda_{\text{identity}} = 0.0$. All primary comparisons in this paper (Sections 5.1, 5.2, 5.4, 5.6) use Track B models, trained under an identical recipe, ensuring a fair comparison across architectures.

The choice of $\lambda_{\text{identity}} = 0.0$ was checked by a hyperparameter sweep across six values (0.0, 0.05, 0.1, 0.2, 0.3, 0.5; MobileNetV2, $n = 5$ seeds each). Every $\lambda_{\text{identity}} > 0$ produced a lower mean downstream accuracy than $\lambda_{\text{identity}} = 0.0$, a consistent direction across all five non-zero values, but the effect only reaches Bonferroni-corrected significance at the largest value tested, $\lambda_{\text{identity}} = 0.5$ (Cohen's $d = -2.17$, $p_{\text{Bonf}} = 0.0416$); $\lambda_{\text{identity}} = 0.2$ reaches a trend level ($p_{\text{Bonf}} = 0.0626$), and the remaining values (0.05, 0.1, 0.3) are non-significant ($d$ $-0.30$ to $-1.48$). This consistent-direction-but-mostly-underpowered pattern supports the decision to exclude the identity-loss term from the main recipe without overstating the strength of any single comparison.

Full hyperparameter and implementation details are given in Section 4.4.

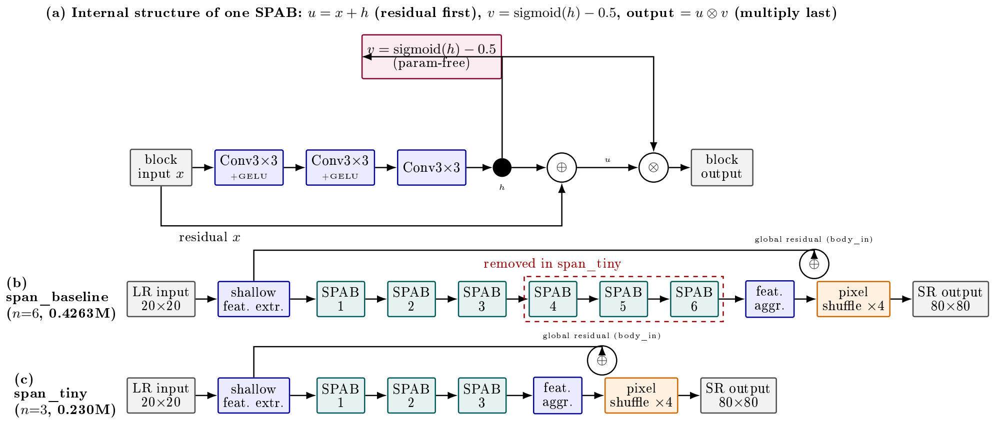



**Fig. 1** SPAN architecture and depth compression, verified directly against the implementation. (a) Internal structure of a single Swift Parameter-free Attention Block (SPAB): three convolutions (first two followed by GELU) produce raw features $h$; $h$ feeds two parallel paths, a residual sum $u = x + h$ and a parameter-free attention map $v = \mathrm{sigmoid}(h) - 0.5$, which are combined by *element-wise multiplication last*, output $= u \otimes v$ (residual add precedes, not follows, the attention multiply). Each SPAB stack also carries a block-level global residual (labelled "global residual (body_in)"): the feature map right after shallow feature extraction is added back after the final feature-aggregation convolution, before pixel-shuffle upsampling. (b) span_baseline (via the official SPAN implementation, `span_official`) chains 6 SPABs. (c) span_tiny removes the last 3 SPABs, reducing deploy-mode parameters by 46.0% (0.4263M → 0.230M) while keeping every other component identical. Note: span_tiny and span_large (Section 5.2) are both built from the same custom `SPAN` re-implementation class shown here (plain 3×3 convolutions), differing only in `n_blocks`; `span_baseline`/`span_official` uses the original authors' reparameterizable Conv3XC design internally, which is why its deploy-mode parameter count (0.4263M) differs slightly from span_large's (0.417M) despite both nominally being 6-block, 48-channel configurations. See the methodological note in Section 3.2.

## 3.4 Recognition backbones

Downstream identity recognition is evaluated on eleven backbone architectures, chosen to span the major design families of lightweight recognition networks rather than a single lineage: hand-designed mobile-efficient CNNs (MobileNetV2 [38], ShuffleNetV2-1.0x [39], SqueezeNet-1.1 [40]), neural-architecture-search-derived designs (EfficientNet-B0 [41], MobileNetV3-Small and MobileNetV3-Large [42]), cheap-operation and design-space-search families (GhostNet-100 [43], RegNetY-400MF [44]), a structural-reparameterization backbone (MobileOne-S0 [45]), a CPU-specific lightweight design (LCNet-100 [46]), and a general-purpose CNN pre-dating the mobile-efficiency era (ResNet-18 [47]). The first five (EfficientNet-B0, GhostNet-100, MobileNetV2, MobileNetV3-Small, ResNet-18) were used from the outset; the remaining six were added subsequently specifically to test whether this paper's findings generalize across architecture families rather than being an artifact of the original five.

## 3.5 Statistical methodology

For every comparison between two SR domains (e.g., span_tiny vs. lr), downstream recognition accuracy is measured across $n = 5$ random seeds (42, 123, 2024, 44, 999), holding the SR model itself fixed. SR models are trained once per configuration, at a fixed seed of 42, and only the downstream recognition training is repeated across seeds for the main result tables; training a full SR model at every recognition seed was judged computationally infeasible at the scale of this study. This leaves open whether the specific SR checkpoint used could itself be an unmeasured source of variance. Section 3.6 measures this directly for the three core architectures. The paired difference in mean accuracy is reported alongside a paired-sample $t$-test (`scipy.stats.ttest_rel`) and a paired effect size (Cohen's $d_z = \text{mean(difference)}/\text{std(difference)}$) [48]. Because multiple backbones and multiple pairwise comparisons are tested within each result table, a Bonferroni correction [49] is applied across the comparisons within each family. $p_{\text{Bonf}} < 0.05$ is adopted as the threshold for a "statistically significant" claim, with $0.05 \leq p_{\text{Bonf}} < 0.10$ distinguished as "suggestive of a trend" and never described as significant in this paper.

$n = 5$ seeds per condition reflects a deliberate breadth-over-depth trade-off under a single-GPU budget (Section 4.4): extending coverage to eleven recognition backbones, two independent datasets, and three AWEx transfer protocols already multiplies the total number of training runs into the thousands, and increasing $n$ per cell further would have required narrowing that coverage substantially instead. At this sample size, a non-significant paired $t$-test alone cannot distinguish a true absence of effect from a real effect the test simply lacks the power to detect. Wherever this paper describes a comparison as safe or accuracy-neutral rather than merely non-significant, that claim is instead backed by an equivalence test: a paired two one-sided tests (TOST) procedure [50, 51], using a pre-specified equivalence margin of $\pm 1$ percentage point of accuracy, chosen to match the informal threshold ("under one percentage point") this paper already uses elsewhere to describe a cost as small. TOST tests, in each one-sided direction, whether the true paired difference exceeds the margin; declaring equivalence requires rejecting both one-sided nulls simultaneously ($p_{\text{TOST}} = \max(p_{\text{lower}}, p_{\text{upper}}) < 0.05$). A comparison can therefore fail to reach significance under the $t$-test above and still not be confirmed equivalent under TOST; this combination is reported as inconclusive rather than folded into either a "significant" or a "safe" verdict.

### 3.6 SR-training-seed variance validation

The multi-seed protocol above holds the SR model fixed at one checkpoint (seed 42) and repeats only the downstream recognition training. To check whether that checkpoint's own training seed is itself a meaningful source of variance (one that the downstream-seed variance already reported would not capture), span_tiny, span_baseline, and span_large were each retrained at four further SR-training seeds (123, 2024, 44, 999), with the downstream recognition seed held fixed at 42 to isolate the SR-training-seed axis from the already-measured downstream-seed axis, across all eleven recognition backbones rather than a single representative one. Table 2 reports, for every backbone and architecture, the ratio of the resulting standard deviation across SR-training seeds to the standard deviation already reported across downstream-recognition seeds for the same configuration.

Two things follow from the full backbone coverage that a single-backbone screen could not have shown. First, the effect is not specific to span_baseline: span_tiny itself

**Table 2** Ratio of the SR-training-seed standard deviation in downstream identity accuracy to the downstream-recognition-seed standard deviation already reported for the same configuration, EarVN1.0, $n$=5 SR-training seeds, all eleven recognition backbones.

| Backbone | span_tiny | span_baseline | span_large |
|---|---|---|---|
| EfficientNet-B0 | 2.44 | 2.43 | 1.10 |
| GhostNet-100 | 1.40 | 1.77 | 2.30 |
| LCNet-100 | 2.74 | 1.01 | 0.70 |
| MobileNetV2 | 1.85 | 2.71 | 1.37 |
| MobileNetV3-Large | 1.28 | 1.50 | 0.97 |
| MobileNetV3-Small | 1.03 | 1.33 | 1.06 |
| MobileOne-S0 | 0.98 | **5.07** | 1.19 |
| RegNetY-400MF | 1.14 | 1.20 | 0.51 |
| ResNet-18 | 1.02 | 1.69 | 0.82 |
| ShuffleNetV2-1.0x | 1.01 | 2.49 | 0.63 |
| SqueezeNet-1.1 | 1.73 | 1.85 | 0.78 |

shows a comparable or larger ratio on three of eleven backbones (EfficientNet-B0, 2.44; LCNet-100, 2.74; MobileNetV2, 1.85), so this is not a limitation isolated to the comparison baseline. Second, the pattern tracks the recognition backbone at least as much as the SR architecture: EfficientNet-B0 shows an elevated ratio for span_tiny and span_baseline alike (but not span_large), while GhostNet-100's highest ratio occurs under span_large, rather than any one SR architecture being uniformly more SR-seed-sensitive than the others. MobileOne-S0 shows the single most extreme value in the table (5.07, span_baseline), well above every other entry, though its span_tiny and span_large ratios for the same backbone are unremarkable (0.98 and 1.19), underscoring that elevated ratios are a backbone$\times$architecture-specific phenomenon rather than a property of either axis alone. Across all 33 backbone$\times$architecture combinations, the SR-training-seed standard deviation exceeds the downstream-recognition-seed standard deviation in 26 of them, most consistently for span_baseline (11/11, i.e. every backbone) and span_tiny (10/11), less consistently for span_large (5/11).

For Table 4's central comparison specifically (span_tiny vs. span_baseline), a fully crossed design was collected for all eleven recognition backbones (all five SR-training seeds against all five downstream-recognition seeds, both domains, a complete 25-cell grid per backbone$\times$domain, $11 \times 25 \times 2 = 550$ observations) and analyzed with a linear mixed-effects model (`statsmodels MixedLM`, REML) with crossed random intercepts for SR-training seed and downstream-recognition seed (each nested within domain, since a shared seed label across span_tiny and span_baseline reflects no shared randomness), in place of the descriptive standard-deviation ratio used above. This separates the two variance sources explicitly instead of comparing two independently estimated standard deviations: the model attributes 20.2% of total variance to the SR-training seed and the remaining 79.8% to residual (backbone- and domain-adjusted) variation; the downstream-recognition-seed component is estimated at 0.0%, pushed fully to the boundary of the parameter space under REML (discussed further below), so unlike the descriptive ratios in Table 2, this model's SR-seed-to-downstream-seed ratio cannot be reported as a precise number here – the qualitative conclusion that SR-training-seed variance is the larger of the two sources is, if anything, reinforced by this boundary estimate, not undermined by it. This is broadly consistent in direction with two earlier-stage fits reported during this analysis: a five-backbone-only fit restricted to the originally-tested backbones (32.2%/1.0%/66.9%, ratio 34:1), and an interim eleven-backbone fit that included the six added backbones with only a sparser 9-cell grid each instead of the full 25 (28.1%/0.6%/71.3%, ratio 44:1, $n = 376$), before their own dedicated core-grid training was completed. All three fits agree on the direction of the effect and on Table 2's per-backbone descriptive ratios (SR-training-seed variance exceeding downstream-seed variance in 26 of 33 backbone/architecture combinations); the complete-grid fit's downstream-seed component landing exactly at the boundary is a more extreme instance of the same boundary-condition warning already present in the earlier fits (the downstream-seed component being pushed toward the edge of the parameter space in all three), so this result should be read as confirming the direction of the effect – not as a precise, or even a well-defined, magnitude.

This variance decomposition motivates a direct methodological concern: every per-backbone significance test reported for the `span_tiny` vs. `span_baseline` comparison in Table 4 (and in every other $n = 5$-seed table in this paper) holds the SR-training seed fixed at 42 and varies only the downstream-recognition seed, so its $p$-value reflects only the downstream-recognition-seed source of variance – which this section shows is the smaller of the two by roughly an order of magnitude. To check whether the paper's central claim survives when both variance sources are modeled jointly instead of one being held fixed, the same crossed model was used to test a linear contrast for the `span_tiny`−`span_baseline` effect averaged equally across all eleven backbones (the model's `condition[T.span_tiny]` fixed-effect coefficient alone is *not* this average – with a `condition`×`backbone` interaction term in the model, that coefficient reflects only the reference backbone's own effect, so the correct pooled test requires the fixed-effect coefficient plus the mean of all eleven interaction coefficients, evaluated via a joint linear-contrast test, not read off the summary table directly). This pooled, variance-correct test gives an estimated effect of $-0.0011$ (SE $= 0.0037$, $p = 0.7759$): no significant difference between `span_tiny` and `span_baseline`, averaged across all eleven backbones, once both the SR-training-seed and downstream-recognition-seed sources of variance are properly accounted for, instead of one being treated as fixed. This directly addresses the concern that Table 4's per-backbone $p$-values might be computed on an incomplete variance source: the central compression-safety claim is not an artifact of that simplification. Non-significance alone ($p = 0.7759$) is not itself evidence of equivalence (Section 3.5), so this pooled estimate was additionally submitted to the same TOST procedure used throughout this paper, against the same $\pm 1$ percentage point margin: both one-sided tests reject non-equivalence (against a true effect $\leq -1$ point, $z = +2.421$, $p = 0.00774$; against a true effect $\geq +1$ point, $z = -2.990$, $p = 0.00139$), giving $p_{\text{TOST}} = \max(0.00774, 0.00139) = 0.00774 < 0.05$. Averaged across all eleven backbones and with both variance sources modeled jointly, span_tiny and span_baseline are therefore confirmed statistically equivalent within $\pm 1$ percentage point – a stronger claim than the per-backbone non-significance results in Table 4 support individually. As with the variance-decomposition estimate above, this pooled model carries the same boundary/convergence caveat, so this TOST result should be read as confirming equivalence at the stated margin and direction rather than as an exact $p$-value.

An initial screening pass restricted to mobilenet_v2 alone had suggested this was a property specific to span_baseline (ratio 3.6× at $n = 3$, 2.71× at $n = 5$), with span_tiny and span_large both close to parity (1.16 and 0.91). Extending the check to all eleven backbones (Table 2) does not support that reading: span_tiny itself reaches a comparable or larger ratio on several backbones, and the architecture with the highest ratio changes depending on which backbone is used for recognition. The pattern is better described as backbone-dependent (EfficientNet-B0 elevated under span_tiny and span_baseline; GhostNet-100 elevated under span_large; MobileOne-S0 elevated under span_baseline specifically, far more so than any other backbone×architecture combination observed) than as a defect specific to any one SR architecture, and no confirmed mechanism is offered for why particular backbone/architecture combinations are more SR-seed-sensitive than others. span_baseline's fixed pretrained initialization (as opposed to span_tiny's and span_large's training from scratch) removes one nominal source of seed-dependence rather than adding one, so a simple weight-initialization account does not explain the pattern either, and it is left as an open question. span_tiny's own point estimates in Table 4 are unaffected; the extra source of variance identified here applies to the pipeline generally, not to the comparison baseline alone, and is discussed further in Section 7.

### 3.7 Data and training integrity validation

During experimentation on the second dataset (AWEx), two training-pipeline issues were identified and corrected by inspecting raw training logs rather than relying solely on aggregated summary statistics, a practice adopted throughout this study after aggregated CSV outputs were found to occasionally mask anomalies visible only in per-epoch logs. First, the `span_official` teacher model exhibited NaN losses during from-scratch training on AWEx (best validation L1 = 0.19, versus ≈0.02–0.03 for healthy runs), traced to numerical overflow under mixed-precision (FP16) training combined with the wrapper's internal `img_range=255` scaling. Disabling automatic mixed precision specifically for this architecture resolved the issue, and PSNR recovered from 9.08 dB to 25.11 dB. Second, a stale teacher checkpoint was inadvertently retained in an auto-generated fine-tuning configuration, causing an initial transfer-learning run to regress relative to its zero-shot starting point. Retraining after the teacher-checkpoint issue was fixed corrected this: fine-tuned PSNR reached 25.135 dB, above the zero-shot value of 24.989 dB, in the expected direction. All results reported in Section 5.6 use the corrected, post-fix data. This validation process is described here both for transparency and because it may offer methodological guidance for others training SPAN-family architectures under mixed precision.

# 4 Experimental Setup

## 4.1 Datasets

**EarVN1.0** (primary dataset) [21]: 164 subjects, used with an automatically-selected low-resolution generation threshold (`hr_source_min=41`, the 20th percentile of the dataset's native resolution distribution). LR images are generated by bicubic downsampling from an 80×80 HR crop to 20×20 (scale ×4).

**AWEx** (secondary dataset, generalization check): 336 subjects, 4,004 images, assembled from three sources released by the same research group (University of Ljubljana): the original AWE set (100 subjects, 1,000 images), the CVL Ear dataset (16 subjects, 804 images), and an additional "New Images" collection (220 subjects, 2,200 images) [22]. All 336 subjects are pooled into a single set with a custom per-subject train/validation/test split (2,260/408/572 images), rather than AWEx's official open-set split (train = AWE+CVL, test = New Images, with disjoint subject identities). The official split is designed for open-set verification, whereas this study addresses closed-set identity classification, matching the protocol used for EarVN1.0. `hr_source_min` is likewise selected automatically (38, the 20th percentile of AWEx's native resolution distribution); the different numeric value relative to EarVN1.0 reflects the two datasets' differing native resolution distributions under an identical selection algorithm, not a protocol inconsistency.

The unmodified AWE dataset (100 subjects, exactly 10 images/subject, ≈5.7 images/subject after splitting) was initially used as the generalization check. Inter-seed variance in downstream accuracy proved comparable in magnitude to the mean effect sizes under study, precluding reliable conclusions even after applying transfer learning and backbone-freezing to reduce variance. The larger, pooled AWEx set was adopted instead, for greater statistical power.

Table 3 reports native-resolution percentiles (short side, pixels) and split sizes for both datasets, measured directly on the full raw image sets before thresholding.

**Table 3** Descriptive statistics for both datasets (native short-side resolution percentiles and split sizes), measured on the full raw image sets.

| Statistic | EarVN1.0 (28,412 images) | AWEx (4,004 images) |
|---|---|---|
| Native short side: min / p5 / p10 | 13 / 27 / 32 px | 12 / 27 / 31 px |
| Native short side: p20 / p25 / p50 | 41 / 46 / 77 px | 38 / 41 / 60 px |
| Native short side: p75 / p90 / max | 122 / 161 / 472 px | 95 / 149 / 557 px |
| `hr_source_min` (p20 threshold) | 41 px (80.85% kept) | 38 px (80.92% kept) |
| `hr_size` / LR size (auto-selected, ×4) | 80 / 20 px | 76 / 19 px |
| `too_small_max` (real-LR holdout) | <20 px (173 images) | <20 px (32 images) |
| Train / validation / test images | 16,077 / 3,415 / 3,480 | 2,260 / 408 / 572 |
| Subjects (identities) | 164 (all represented in every split) | 336 (all represented in every split) |
| Gender distribution (subjects) | 98 male / 66 female | not available (see note below) |

Gender labels for EarVN1.0 follow the folder-range convention documented by the dataset's original authors (subjects 1–98 male, 99–164 female) [21], and the resulting 98/66 split above is computed directly from `splits.json` (consistent across all three splits: 62.1%/37.9% in train, 62.0%/38.0% in val, 62.1%/37.9% in test).

AWEx carries no equivalent ground truth: it is assembled from three source collections whose subjects are renumbered by directory-listing order during preprocessing (Section 4.1), an ordering with no relationship to gender, so no gender distribution or gender-classification result is reported for AWEx anywhere in this paper.

## 4.2 Evaluation protocol

SR image quality is evaluated with PSNR and SSIM [52] computed on the ear region of interest (ROI) only, and with LPIPS [53] (using ImageNet-normalized features), against a consistent bicubic-upsampling protocol. Rank-1 identity accuracy is the primary metric reported throughout this paper; rank-5, AUC, and EER are computed for every evaluation and reported for Table 4's central comparison in Table 30 (Supplementary Material), while full confusion matrices are retained for error analysis but not presented here. Gender-classification accuracy is reported where the underlying dataset provides real gender ground truth (Section 4.1).

## 4.3 Baselines

The following SR configurations are compared: `edsr_teacher` (an EDSR-based [54] upper-bound reference), `span_baseline`/`span_official` (uncompressed SPAN), `span_large` (depth-ablation counterpart to span_tiny), and, in the contextual comparison of Sections 5.4/5.6, `rlfn`, `rlfn_adapted`, `ecbsr`, `safmn`, and `smfanet`, each trained under both Track A and Track B where applicable. `rlfn_adapted` uses a modified ESA-pooling kernel/stride configuration (kernel=3, stride=2) rather than the original authors' setting (kernel=7, stride=3), because at 20×20 input resolution the original configuration was found to degenerate toward a near-channel-attention mechanism. `rlfn_adapted` is therefore used as the primary RLFN baseline throughout, with the unmodified `rlfn` reported only as a secondary reference.

## 4.4 Implementation details

All models are trained with Adam [55] at a fixed learning rate (no scheduler), relying on early stopping (patience 10 epochs on the relevant validation metric, maximum 500 epochs) rather than a decay schedule to control convergence. Batch size is 64 throughout. Learning rates differ by training stage: $1 \times 10^{-3}$ for the uncompressed SPAN teacher and other from-scratch Track-A/Track-B SR baselines; $1 \times 10^{-4}$ for span_tiny's distillation recipe and for learned block pruning (with a further 10× reduction for the post-harden fine-tuning stage, Section 5.7); $3 \times 10^{-4}$ (weight decay $5 \times 10^{-4}$) for all recognition-backbone training. Automatic mixed precision is used for recognition training on CUDA, but deliberately *not* for SR training, after the FP16 numerical-overflow issue described in Section 3.7. Data augmentation is deliberately minimal: LR-domain images are bicubically resized to the target resolution (a no-op for the `hr`/`sr_baseline`/`sr_improved` domains, whose images are already exported at the correct size); no random cropping, color jitter, or flipping is applied. A random-horizontal-flip augmentation used in earlier development was deliberately removed, since flipping an ear image can turn a left-ear morphology into one resembling a right ear. Ears, unlike faces, are not left-right symmetric [19], making this augmentation biometrically invalid for this modality rather than a free source of extra training diversity.

All eleven recognition backbones are initialized from ImageNet-pretrained weights (via `torchvision` for nine backbones, `timm` for GhostNet-100/MobileOne-S0/LCNet-100) rather than trained from scratch, following the transfer-learning necessity Alshazly et al. [13] report for ear recognition given the small size of available ear datasets.

All experiments were run with PyTorch 2.11.0 (CUDA 13.0, cuDNN 9.19), torchvision 0.26.0, and timm 1.0.26, on a single NVIDIA GeForce RTX 3080 (10 GB, compute capability 8.6).

# 5 Results

Before turning to the individual comparisons, it is worth stating explicitly what these accuracy numbers mean in absolute terms, since a reader unfamiliar with this specific regime may otherwise read rank-1 accuracy in the 0.40–0.56 range on EarVN1.0 (Table 4) and 0.09–0.21 on AWEx (Section 5.6) as surprisingly low. Both figures reflect the difficulty of the task itself rather than a weakness of any SR method tested here: recognition is closed-set identity classification over 164 subjects (EarVN1.0) or 336 subjects (AWEx, Section 4.1) from a $20 \times 20$-pixel input (Section 3), roughly an order of magnitude below the input resolution typical face- or ear-recognition benchmarks use, and rank-1 accuracy over this many closed-set classes at this resolution is inherently far from the near-ceiling numbers such benchmarks report. Table 5 makes the point directly: even given the true, uncompressed high-resolution image – no super-resolution step at all – the achievable ceiling on EarVN1.0 tops out at 0.58–0.74 across backbones, and span_tiny closes only 12.5–25.9% of the gap between no-SR and that ceiling (Section 5.1.1). The absolute accuracy level is therefore a property of the resolution/identity-count regime under study, not a deficiency of the compression method evaluated here; this paper's claims throughout concern the *relative* ordering and cost among SR domains at this fixed, extreme resolution, not the absolute accuracy achievable at it.

## 5.1 Main result (EarVN1.0): accuracy ordering across eleven recognition backbones

Table 4 reports downstream rank-1 identity accuracy for the `lr` (no-SR), `span_tiny`, and `span_baseline` domains across all eleven recognition backbones (the original five plus six added afterward to test generality across architecture families), together with paired statistical tests for the two comparisons central to this paper's claim.

`span_tiny` > `lr` reaches statistical significance ($p_{\mathrm{Bonf}} < 0.05$) in all eleven backbones, with large effect sizes throughout (Cohen's $d$ 2.1–9.1). `span_baseline` > `lr` is significant in all eleven backbones as well ($d$ 2.6–10.6, not tabulated separately). The accuracy cost of compression itself, `span_tiny` vs. `span_baseline`, reaches significance in only 1/11 backbones (EfficientNet-B0, a 1.2-percentage-point cost); the other 10 are statistically indistinguishable, with span_tiny fractionally *higher* than the baseline in four of them (MobileNetV3-Large, MobileNetV3-Small, MobileOne-S0,

**Table 4** Downstream identity accuracy (rank-1) on EarVN1.0, $n = 5$ seeds. "tiny>lr" and "tiny vs. baseline" report paired Cohen's $d$ and Bonferroni-corrected $p$-values (family of 3 comparisons per backbone: lr–baseline, lr–tiny, baseline–tiny). "TOST" is the equivalence test for tiny vs. baseline (Section 3.5, ±1 percentage point margin); "not equiv." means equivalence is not confirmed at $n = 5$, which is a distinct claim from "significant"; see accompanying text.

| Backbone | lr | span_tiny | span_baseline | tiny > lr ($d$, $p_{\mathrm{Bonf}}$) | tiny vs. baseline ($d$, $p_{\mathrm{Bonf}}$) | TOST ($p_{\mathrm{TOST}}$) |
|---|---|---|---|---|---|---|
| EfficientNet-B0 | 0.5125 | 0.5485 | 0.5604 | $d$=8.38, $p$=0.0001 (sig.) | $d$=−2.45, $p$=0.0163 (sig., tiny lower) | 0.79 (not equiv.) |
| GhostNet-100 | 0.4675 | 0.5115 | 0.5205 | $d$=9.10, $p$=0.0001 (sig.) | $d$=−1.36, $p$=0.1163 (n.s.) | 0.38 (not equiv.) |
| LCNet-100 | 0.4279 | 0.4852 | 0.5018 | $d$=3.92, $p$=0.0028 (sig.) | $d$=−1.15, $p$=0.1857 (n.s.) | 0.82 (not equiv.) |
| MobileNetV2 | 0.4444 | 0.5026 | 0.5125 | $d$=7.75, $p$=0.0002 (sig.) | $d$=−0.74, $p$=0.5135 (n.s.) | 0.50 (not equiv.) |
| MobileNetV3-Large | 0.4492 | 0.5072 | 0.5052 | $d$=2.50, $p$=0.0150 (sig.) | $d$=0.17, $p$=1.0 (n.s., tiny higher) | 0.09 (not equiv.) |
| MobileNetV3-Small | 0.4002 | 0.4423 | 0.4398 | $d$=4.41, $p$=0.0018 (sig.) | $d$=0.27, $p$=1.0 (n.s., tiny higher) | 0.07 (not equiv.) |
| MobileOne-S0 | 0.4550 | 0.4898 | 0.4852 | $d$=2.12, $p$=0.0272 (sig.) | $d$=0.41, $p$=1.0 (n.s., tiny higher) | 0.17 (not equiv.) |
| RegNetY-400MF | 0.4530 | 0.5107 | 0.5210 | $d$=6.34, $p$=0.0004 (sig.) | $d$=−0.68, $p$=0.6058 (n.s.) | 0.52 (not equiv.) |
| ResNet-18 | 0.4575 | 0.4825 | 0.4882 | $d$=2.20, $p$=0.0239 (sig.) | $d$=−0.66, $p$=0.6379 (n.s.) | 0.16 (not equiv.) |
| ShuffleNetV2-1.0x | 0.4451 | 0.4928 | 0.4907 | $d$=5.58, $p$=0.0007 (sig.) | $d$=0.28, $p$=1.0 (n.s., tiny higher) | 0.04 (equiv.) |
| SqueezeNet-1.1 | 0.4011 | 0.4352 | 0.4416 | $d$=3.63, $p$=0.0038 (sig.) | $d$=−0.65, $p$=0.6545 (n.s.) | 0.23 (not equiv.) |

ShuffleNetV2-1.0x). Each of these 10 was additionally tested for equivalence (TOST, ±1 percentage point margin, Section 3.5): one reaches the threshold for confirmed equivalence at $n = 5$ seeds, ShuffleNetV2-1.0x ($p_{\mathrm{TOST}}$=0.04); the remaining nine do not ($p_{\mathrm{TOST}}$ 0.07–0.82), MobileNetV3-Small coming closest among them. This paper therefore reports the compression cost as *not detected as significant* in ten of eleven backbones (a real and useful finding, since it means no measured evidence points to a cost there) and *confirmed accuracy-neutral* in one of those ten; the depth ablation below (Table 6), which isolates the same question with a cleaner single-variable comparison, provides a further, methodologically cleaner check of the same question.

### 5.1.1 How much of the resolution gap does SR actually close?

Table 4 establishes that SR helps relative to no-SR and that compression costs little relative to the uncompressed baseline, but neither comparison says how much headroom remains: what is the best downstream accuracy achievable if resolution itself were not a limitation? To answer this, each backbone was additionally trained and evaluated directly on the true high-resolution (HR) crops (no super-resolution step at all), at the same $n$=5-seed protocol as every other domain in Table 4 – a theoretical ceiling rather than a competing method. Table 5 reports it; $p_{\mathrm{Bonf}}$ here is corrected across a 6-comparison family per backbone ($\binom{4}{2}$ over hr/lr/span_baseline/span_tiny), distinct from Table 4's 3-comparison family, since this is a separate four-domain analysis that does not alter Table 4's own figures.

The gap is large and consistent: on all eleven backbones, the HR ceiling sits roughly 18–28 percentage points above `lr`, and `span_tiny` closes only 12.5–25.9% of it (mean 20.4%), `span_baseline` closing a similar 15.4–30.1% (mean 22.6%) – the two track each other closely throughout, consistent with the paper's central claim that compression costs little, since both start from the same modest fraction of the available headroom. The hr-vs-span_tiny gap is significant after Bonferroni correction in 11 of 11 backbones (as is hr-vs-span_baseline, $d$ ranging from −5.86 to −20.84), confirming the ceiling is real and not an artifact of measurement noise. The largest effect, ResNet-18 at $d$=−46.83, is a paired-sample $d$ specifically, not evidence of an unusually reliable or different result: `hr` and `span_tiny` are trained with the same 5 seeds, so

**Table 5** HR ceiling vs. the three reconstructed domains, downstream identity accuracy, EarVN1.0, $n = 5$ seeds, all eleven recognition backbones. "% closed" is $(\text{domain} - \texttt{lr})/(\text{hr} - \texttt{lr}) \times 100$, the share of the no-SR-to-ceiling gap recovered by that domain. The last column is the hr-vs-span_tiny comparison, Bonferroni-corrected over the 6-comparison family; $d$ is the paired-sample effect size and is often large throughout this table (rather than isolated to one row) because `hr` and the reconstructed domains share the same 5 seeds, so seed-driven variation moves them together and the paired-difference standard deviation is typically much smaller than either domain's own standard deviation – discussed further in the text for the most extreme case.

| Backbone | hr (ceiling) | lr | span_baseline | span_tiny | % closed (baseline) | % closed (tiny) | hr vs. tiny ($d$, $p_{\text{Bonf}}$) |
|---|---|---|---|---|---|---|---|
| EfficientNet-B0 | 0.7421 | 0.5125 | 0.5604 | 0.5485 | 20.9% | 15.7% | $d$=−17.73, $p$<0.0001 |
| GhostNet-100 | 0.7041 | 0.4675 | 0.5205 | 0.5115 | 22.4% | 18.6% | $d$=−10.97, $p$=0.0001 |
| LCNet-100 | 0.6734 | 0.4279 | 0.5018 | 0.4852 | 30.1% | 23.3% | $d$=−12.08, $p$=0.0001 |
| MobileNetV2 | 0.6876 | 0.4444 | 0.5125 | 0.5026 | 28.0% | 23.9% | $d$=−13.33, $p$<0.0001 |
| MobileNetV3-Large | 0.7332 | 0.4492 | 0.5052 | 0.5072 | 19.7% | 20.4% | $d$=−4.16, $p$=0.0045 |
| MobileNetV3-Small | 0.5795 | 0.4002 | 0.4398 | 0.4423 | 22.1% | 23.5% | $d$=−2.71, $p$=0.0225 |
| MobileOne-S0 | 0.6486 | 0.4550 | 0.4852 | 0.4898 | 15.6% | 18.0% | $d$=−4.08, $p$=0.0048 |
| RegNetY-400MF | 0.6913 | 0.4530 | 0.5210 | 0.5107 | 28.5% | 24.2% | $d$=−17.94, $p$<0.0001 |
| ResNet-18 | 0.6570 | 0.4575 | 0.4882 | 0.4825 | 15.4% | 12.5% | $d$=−46.83, $p$<0.0001 |
| ShuffleNetV2-1.0x | 0.6294 | 0.4451 | 0.4907 | 0.4928 | 24.7% | 25.9% | $d$=−6.44, $p$=0.0008 |
| SqueezeNet-1.1 | 0.5922 | 0.4011 | 0.4416 | 0.4352 | 21.2% | 17.8% | $d$=−13.39, $p$<0.0001 |

seed-driven variation (data split, initialization) moves both similarly, leaving an especially small paired-difference standard deviation for this backbone (0.37 percentage points) relative to the size of the gap itself (17.45 points, 95% CI $[16.99, 17.91]$) – the confidence interval is in fact among the tightest in the table, not the widest, which is the more informative way to read this row than the effect size alone.

This result directly answers a question the paper's other comparisons do not: SR recovers a real but modest fraction, roughly a fifth to a quarter, of the accuracy that genuine full resolution would provide, across all eleven backbones tested. This does not bear on the paper's central claim, which concerns the relative cost of *compressing* the SR model, not the absolute gap between any SR reconstruction and true high resolution – closing that larger gap is a harder problem this paper's compression question does not attempt to solve.

Figure 2 shows qualitative examples on real EarVN1.0 test images. Across the full 20-sample export, the per-image PSNR gap (span_baseline − span_tiny) ranges from −0.062 dB, with span_tiny marginally ahead (Figure 2a), to +2.645 dB, the largest gap observed (Figure 2e). Samples (a)–(e) are selected to span this full range rather than to favor either model. A follow-up export at $n = 200$ samples, an order of magnitude larger and independently drawn (not a superset of the original 20), confirms this range is not an artifact of a small, unrepresentative sample: the largest gap unfavorable to span_tiny across all 200 samples is +1.19 dB, smaller in magnitude than the +2.645 dB case already shown here, and the single largest gap of either sign across the 200 samples in fact favors span_tiny (−2.251 dB). Two patterns are visible. First, the visual gap between span_baseline and span_tiny tracks the PSNR gap: it is imperceptible when the two are close (Figure 2a–b) and visible mainly as reduced fine-structure sharpness in the antihelix/concha region when the gap is largest (Figure 2e), consistent with a small, non-catastrophic accuracy cost rather than a qualitatively different failure mode. Second, Figure 2b illustrates a limitation of pixel-fidelity metrics noted in Section 2: the ear itself is largely occluded by hair in this sample, so the near-tied, relatively high PSNR (30.92 vs. 30.89 dB) is driven substantially by accurate reconstruction of the large, low-detail hair and background regions rather than by ear-specific structure. PSNR and downstream identity accuracy are related but not interchangeable signals.

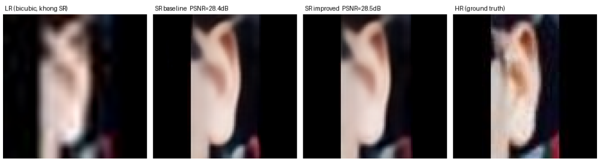
These per-image values are computed over the full image frame, including any letterbox padding, whereas the dataset-level averages in Table 10 are computed on the ear region of interest only (Section 4). The two are therefore not directly comparable in absolute terms; the same distinction applies to the AWEx qualitative comparison (Figure 4).

## 5.2 Ablation: depth (span_tiny vs. span_large)

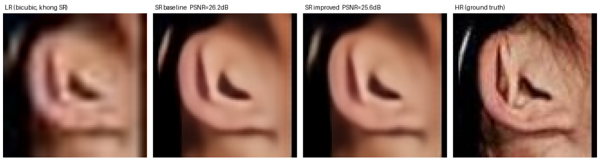
**Table 6** span_tiny vs. span_large, downstream identity accuracy, EarVN1.0, $n = 5$ seeds, all eleven recognition backbones. Difference and $d$ are span_large − span_tiny; $p_{\mathrm{Bonf}}$ corrected across the 6-comparison family reported by this ablation's own aggregation (Section 3.5). $p_{\mathrm{TOST}}$ is the equivalence test (±1 percentage point margin); "equiv." requires $p_{\mathrm{TOST}} < 0.05$, a stricter and distinct claim from "n.s."

| Backbone | span_tiny | span_large | Difference | Cohen's $d$ | $p_{\mathrm{Bonf}}$ | $p_{\mathrm{TOST}}$ |
|---|---|---|---|---|---|---|
| EfficientNet-B0 | 0.5485 | 0.5568 | +0.0083 | 3.17 | 0.0125 (sig.) | 0.11 (not equiv.) |
| GhostNet-100 | 0.5115 | 0.5140 | +0.0025 | 1.33 | 0.244 | **0.0004 (equiv.)** |
| LCNet-100 | 0.4852 | 0.4966 | +0.0114 | 0.65 | 1.0 | 0.57 (not equiv.) |
| MobileNetV2 | 0.5026 | 0.5044 | +0.0018 | 0.29 | 1.0 | **0.0235 (equiv.)** |
| MobileNetV3-Large | 0.5072 | 0.5171 | +0.0099 | 0.41 | 1.0 | 0.50 (not equiv.) |
| MobileNetV3-Small | 0.4423 | 0.4473 | +0.0050 | 0.37 | 1.0 | 0.23 (not equiv.) |
| MobileOne-S0 | 0.4898 | 0.4837 | −0.0060 | −0.59 | 1.0 | 0.22 (not equiv.) |
| RegNetY-400MF | 0.5107 | 0.5147 | +0.0040 | 0.54 | 1.0 | 0.07 (not equiv.) |
| ResNet-18 | 0.4825 | 0.4863 | +0.0038 | 0.45 | 1.0 | 0.09 (not equiv.) |
| ShuffleNetV2-1.0x | 0.4928 | 0.4923 | −0.0005 | −0.03 | 1.0 | 0.13 (not equiv.) |
| SqueezeNet-1.1 | 0.4352 | 0.4330 | −0.0021 | −0.48 | 1.0 | **0.0087 (equiv.)** |

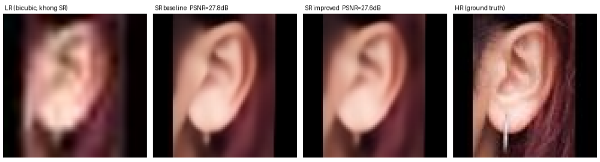
10/11 backbones show no statistically significant difference between span_tiny (0.230M parameters) and span_large (0.417M parameters). The exception, EfficientNet-B0, shows a small but statistically significant cost ($p_{\mathrm{Bonf}}$=0.0125), span_large ahead by 0.83 percentage points, and this cost is also not confirmed within the equivalence margin by TOST ($p_{\mathrm{TOST}}$=0.11), a consistent, real (if small) cost on this backbone by both tests. Of the remaining ten, TOST confirms genuine equivalence within ±1 percentage point for three (GhostNet-100, $p_{\mathrm{TOST}}$=0.0004; MobileNetV2, $p_{\mathrm{TOST}}$=0.0235; SqueezeNet-1.1, $p_{\mathrm{TOST}}$=0.0087); the other seven (LCNet-100, MobileNetV3-Large, MobileNetV3-Small, MobileOne-S0, RegNetY-400MF, ResNet-18, ShuffleNetV2-1.0x) are non-significant under the paired $t$-test but do not reach the TOST threshold either, and are reported as inconclusive. The paper's central claim is therefore precisely this: reducing depth from 6 to 3 SPABs is confirmed accuracy-neutral in 3/11 backbones, shows a small but real cost in 1/11, and remains an open question at this sample size in the remaining 7/11, not a blanket claim of zero cost everywhere, and stronger where it is made than a non-significance argument alone would support. On AWEx (Section 5.6), the same comparison is statistically non-significant in 11/11 backbones, though at a smaller sample for the original five ($n = 3$ seeds) and the standard $n = 5$ for the six added later; see Section 7.



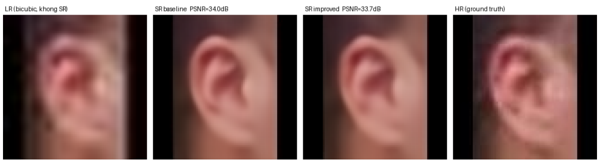

(a) baseline 28.654 dB/SSIM 0.8968, span_tiny 28.716 dB/SSIM 0.8824: span_tiny marginally ahead on PSNR




(b) baseline 30.920 dB/SSIM 0.8580, span_tiny 30.890 dB/SSIM 0.8544: near-tie, ear largely occluded by hair




(c) baseline 25.656 dB/SSIM 0.8576, span_tiny 25.252 dB/SSIM 0.8395: lowest-PSNR pair in the 20-sample export




(d) baseline 34.946 dB/SSIM 0.9067, span_tiny 33.716 dB/SSIM 0.8908: highest-PSNR pair in the 20-sample export


(e) baseline 31.413 dB/SSIM 0.9075, span_tiny 28.768 dB/SSIM 0.8813: largest observed gap across all 20 exported EarVN1.0 samples

**Fig. 2** Qualitative comparison on five EarVN1.0 test images, each showing (left to right) the bicubic-upsampled LR input (no SR), span_baseline output, span_tiny output, and the HR ground truth. Samples (a)–(e) were selected from a 20-sample export to span the full observed per-image PSNR gap, from span_tiny marginally ahead (a) to the largest gap in favor of span_baseline (e), together with the lowest- and highest-PSNR pairs observed (c, d), using the exact logged values rather than a curated set of favorable examples.

### 5.2.1 Locating the compression boundary: a full depth sweep

The comparison above isolates a single point on the depth axis (3 vs. 6 blocks) and therefore cannot say how far compression can be pushed before it costs accuracy, only that this particular cut does not. To answer that question directly, span_tiny's two intermediate block counts (1, 2, 4, 5) were each trained under the identical pinned recipe used for span_tiny itself (Section 3.3, isolating block count as the only varying factor) and evaluated at $n = 5$ seeds on MobileNetV2, the representative backbone used for other screening-scale checks in this study, then extended to four further backbones spanning distinct architecture families (ResNet-18, EfficientNet-B0, GhostNet-100, ShuffleNetV2-1.0x) to test whether a compression boundary identified on a single screening backbone generalizes. Figure 3 plots all five backbones' six-point curves together, Table 7 reports the same values numerically, and Table 8 compares each compressed configuration against the uncompressed 6-block reference, backbone by backbone.

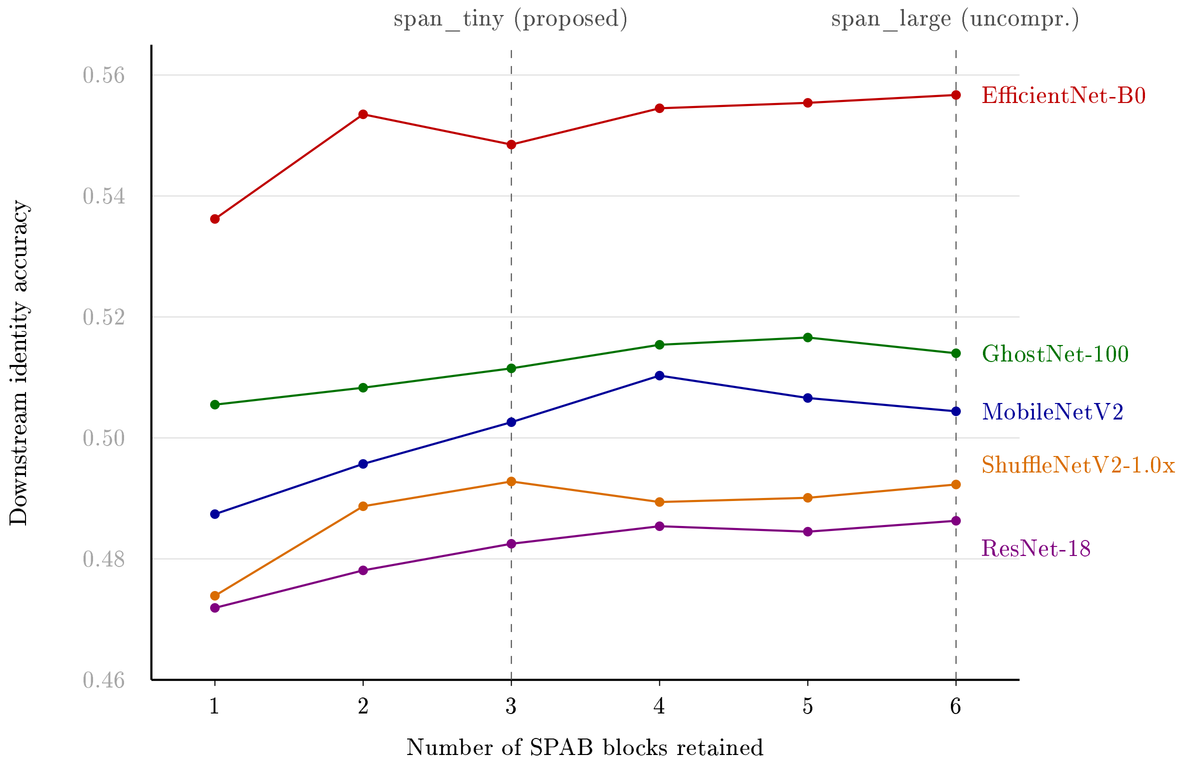



**Fig. 3** Downstream identity accuracy as a function of retained SPAB block count, EarVN1.0, $n = 5$ seeds, five backbones (mean values; std. dev. and per-point TOST verdicts in Table 7 and Table 8). Note the non-monotonic dip for EfficientNet-B0 at exactly 3 blocks (span_tiny's own choice), and the comparatively flat, low-slope curves for ResNet-18 and ShuffleNetV2-1.0x, for which no block count in the 2–5 range is TOST-confirmed equivalent to the 6-block reference.

**Table 7** Downstream identity accuracy (mean ± std. dev.) across block counts 1–6, EarVN1.0, $n = 5$ seeds, five backbones (MobileNetV2, the original screening backbone, plus four added to test generality across architecture families). 3 blocks = span_tiny, 6 blocks = span_large; both reproduced from Table 6.

| Backbone | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|
| MobileNetV2 | 0.4874 ± 0.0025 | 0.4957 ± 0.0073 | 0.5026 ± 0.0074 | 0.5103 ± 0.0089 | 0.5066 ± 0.0032 | 0.5044 ± 0.0048 |
| EfficientNet-B0 | 0.5362 ± 0.0071 | 0.5535 ± 0.0100 | 0.5485 ± 0.0035 | 0.5545 ± 0.0018 | 0.5554 ± 0.0078 | 0.5567 ± 0.0043 |
| GhostNet-100 | 0.5055 ± 0.0084 | 0.5083 ± 0.0065 | 0.5115 ± 0.0025 | 0.5154 ± 0.0050 | 0.5166 ± 0.0057 | 0.5140 ± 0.0036 |
| ResNet-18 | 0.4719 ± 0.0053 | 0.4781 ± 0.0095 | 0.4825 ± 0.0065 | 0.4854 ± 0.0090 | 0.4845 ± 0.0097 | 0.4863 ± 0.0066 |
| ShuffleNetV2-1.0x | 0.4739 ± 0.0058 | 0.4887 ± 0.0047 | 0.4928 ± 0.0080 | 0.4894 ± 0.0028 | 0.4901 ± 0.0058 | 0.4923 ± 0.0109 |

**Table 8** Verdict summary: each block count vs. the 6-block (uncompressed) reference, EarVN1.0, $n = 5$ seeds, five backbones. "equiv." is TOST-confirmed equivalence within ±1 percentage point; "inconclusive" is non-significant under the paired $t$-test but not TOST-confirmed equivalent either – a distinct, weaker claim than "equiv." (Section 3.5). Full effect sizes and $p$-values in Table 29 (Supplementary Material).

| Backbone | 1 vs. 6 | 2 vs. 6 | 3 vs. 6 | 4 vs. 6 | 5 vs. 6 |
|---|---|---|---|---|---|
| MobileNetV2 | **sig. cost** | inconclusive | equiv. | inconclusive | equiv. |
| EfficientNet-B0 | **sig. cost** | inconclusive | **sig. cost** | equiv. | equiv. |
| GhostNet-100 | inconclusive | inconclusive | equiv. | equiv. | equiv. |
| ResNet-18 | inconclusive | inconclusive | inconclusive | inconclusive | inconclusive |
| ShuffleNetV2-1.0x | **sig. cost** | inconclusive | inconclusive | inconclusive | inconclusive |

The curve is not monotonic in the range that matters, and this holds across backbones, not only MobileNetV2. Only the most aggressive cut tested, to a single block, produces a large cost that reaches significance after Bonferroni correction in three of the five backbones (MobileNetV2, $d$=−4.59; EfficientNet-B0, $d$=−2.54; ShuffleNetV2-1.0x, $d$=−2.68; all $p_{\mathrm{Bonf}}$<0.05), while GhostNet-100 and ResNet-18 show no significant cost even at this most extreme cut once corrected. EfficientNet-B0's curve is additionally non-monotonic: the 3-block cut (span_tiny's own choice) shows a significant cost ($d$=−3.17, $p_{\mathrm{Bonf}}$=0.0104) while the *less*-compressed 2-block point does not ($d$=−0.28, n.s.), and TOST confirms equivalence again at both 4 and 5 blocks. This is not a clean monotonic decline that recovers with more capacity; it is a local dip at exactly span_tiny's chosen depth for this one backbone, consistent with EfficientNet-B0 already being the sole exception in Table 6's single-point comparison. GhostNet-100 is the most permissive of the five: no block count tested shows a significant cost, and TOST confirms equivalence to the uncompressed reference from 3 blocks onward. ResNet-18 and ShuffleNetV2-1.0x sit at the opposite end: beyond the single-block extreme (significant for ShuffleNetV2-1.0x only), every intermediate point for both backbones is non-significant *and* not TOST-confirmed equivalent – the sweep identifies no confirmed floor of safe compression for either backbone at this sample size, only the absence of detected harm, which is a weaker claim. The confirmed floor of safe compression this study can establish is therefore backbone-dependent rather than a single number: 3 blocks for MobileNetV2 and GhostNet-100 (span_tiny's own choice, TOST-confirmed equivalent for both), an open question across the entire 2–5 block range for ResNet-18 and ShuffleNetV2-1.0x, and a backbone where the 3-block point specifically underperforms relative to both fewer and more blocks for EfficientNet-B0.

This does not change the paper's proposed configuration, since span_tiny's 3-block design has additionally been validated across all eleven backbones and two datasets (Table 6, Section 5.6); it does show that a full depth curve, even extended across five architecturally distinct backbones, does not settle the compression-boundary question in general – the same $n = 5$ statistical-power limitation documented throughout this paper (Section 7) applies here too, and answering it more completely would require more seeds per backbone rather than more backbones alone.

### 5.3 Ablation: loss components

**Table 9** Downstream accuracy, all pairwise comparisons among four training-loss configurations, EarVN1.0, $n = 5$ seeds, all eleven recognition backbones. Difference convention: "A vs. B" = B minus A; $p_{\mathrm{Bonf}}$ corrected across all 6 comparisons within each backbone. sig. = $p_{\mathrm{Bonf}}<0.05$; trend = $0.05 \leq p_{\mathrm{Bonf}}<0.10$; n.s. = $p_{\mathrm{Bonf}} \geq 0.10$. "(equiv.)" indicates TOST-confirmed equivalence within ±1 percentage point.

| Backbone | full vs. id. | full vs. dist. | full vs. only | id. vs. dist. | id. vs. only | dist. vs. only |
|---|---|---|---|---|---|---|
| EfficientNet-B0 | $d$=−1.01, n.s. | $d$=+0.33, n.s. | $d$=+0.19, n.s. | $d$=+1.79, trend | $d$=+1.34, n.s. | $d$=−0.04, n.s. (equiv.) |
| GhostNet-100 | $d$=−0.06, n.s. | $d$=+1.03, n.s. | $d$=+0.83, n.s. | $d$=+0.85, n.s. | $d$=+0.62, n.s. | $d$=−0.16, n.s. (equiv.) |
| LCNet-100 | $d$=+0.27, n.s. | $d$=+0.59, n.s. | $d$=+1.03, n.s. | $d$=+0.21, n.s. | $d$=+2.18, **sig.** | $d$=+0.56, n.s. |
| MobileNetV2 | $d$=−0.84, n.s. | $d$=+1.93, trend | $d$=+3.13, **sig.** | $d$=+2.85, **sig.** | $d$=+2.61, **sig.** | $d$=−0.40, n.s. |
| MobileNetV3-Large | $d$=−0.15, n.s. | $d$=−0.64, n.s. | $d$=−0.25, n.s. | $d$=−0.39, n.s. | $d$=−0.20, n.s. | $d$=+0.36, n.s. |
| MobileNetV3-Small | $d$=+0.40, n.s. | $d$=+0.36, n.s. | $d$=+0.31, n.s. | $d$=−0.05, n.s. | $d$=−0.15, n.s. | $d$=−0.14, n.s. |
| MobileOne-S0 | $d$=−0.51, n.s. | $d$=+0.86, n.s. | $d$=−0.01, n.s. (equiv.) | $d$=+0.81, n.s. | $d$=+0.54, n.s. | $d$=−0.91, n.s. |
| RegNetY-400MF | $d$=−0.55, n.s. | $d$=+0.84, n.s. | $d$=+1.12, n.s. | $d$=+1.59, n.s. | $d$=+1.27, n.s. | $d$=−0.08, n.s. (equiv.) |
| ResNet-18 | $d$=−1.06, n.s. | $d$=−0.24, n.s. (equiv.) | $d$=−0.07, n.s. | $d$=+1.18, n.s. | $d$=+1.35, n.s. | $d$=+0.05, n.s. |
| ShuffleNetV2-1.0x | $d$=+0.10, n.s. | $d$=+1.92, trend | $d$=+2.14, trend | $d$=+1.63, n.s. | $d$=+0.75, n.s. | $d$=−0.42, n.s. |
| SqueezeNet-1.1 | $d$=−0.04, n.s. | $d$=+1.00, n.s. | $d$=+1.11, n.s. | $d$=+0.45, n.s. | $d$=+0.64, n.s. | $d$=+0.65, n.s. (equiv.) |

On MobileNetV2, the six configurations resolve into two statistically-tied clusters: {pixel+distill, pixel-only} both significantly or near-significantly outperform {pixel+distill+identity, pixel+identity} (3/4 cross-cluster comparisons significant, 1/4 at trend level), while the two within-cluster comparisons are both non-significant. Extending this ablation to ten further backbones shows this pattern is backbone-dependent rather than universal, but not unique to MobileNetV2 either. Two backbones echo the same direction: ShuffleNetV2-1.0x reaches trend level on the same two cross-cluster comparisons (full vs. distill, full vs. only), and LCNet-100 reaches significance on one (id. vs. only). The remaining eight backbones (EfficientNet-B0, GhostNet-100, MobileNetV3-Large, MobileNetV3-Small, MobileOne-S0, RegNetY-400MF, ResNet-18, SqueezeNet-1.1) show no comparison reaching even trend level, with six TOST-confirmed equivalences scattered among their non-significant comparisons. The standalone $\lambda_{\mathrm{identity}}$ sweep (Section 3.3, MobileNetV2-only) shows the same consistent negative direction found for MobileNetV2 above. Read together, the identity-loss term reliably hurts downstream accuracy on MobileNetV2 and, more weakly, on two of the ten other backbones tested (one at trend level only), and shows no detectable effect on the remaining eight; this is reported as a genuinely backbone-dependent effect rather than either a general property of the identity-loss mechanism or an artifact isolated to a single backbone. By contrast, the distillation term's own marginal contribution (pixel+distill vs. pixel-only) remains statistically indistinguishable from pixel-only training on all eleven backbones, and is TOST-confirmed equivalent on four of them (EfficientNet-B0, GhostNet-100, RegNetY-400MF, SqueezeNet-1.1); this specific negative result is unchanged and, if anything, more consistently supported than before.

## 5.4 Contextual comparison with other lightweight SR architectures

Table 10 reports SR image-quality metrics for all evaluated architectures, at $n = 5$ SR-training seeds for the three core SPAN-family variants and $n = 1$ for the rest (Section 7). Table 11 reports downstream accuracy comparisons against span_tiny under the shared Track B distillation recipe, all eleven recognition backbones, $n = 5$ seeds.

**Table 10** SR image quality on EarVN1.0 (ROI-only, ×4). span_baseline/span_tiny/span_large, EDSR, and the four Track-B-distilled contextual architectures (RLFN-adapted, ECBSR, SAFMN, SMFANet) report mean±std across the same $n = 5$ SR-training seeds (42, 123, 2024, 44, 999) used for the SR-seed-variance analysis (Section 3.6); the Track-A and half-depth rows remain $n = 1$ (single training run per configuration, not yet extended to multiple SR-training seeds, Section 7). MACs/FLOPs are hardware-independent complexity measures (FLOPs = 2×MACs), computed once per architecture at a fixed $20 \times 20$ input resolution and therefore identical across seeds.

| Model | PSNR (dB) | SSIM | LPIPS↓ | Params (M, deploy) | MACs (G) | FLOPs (G) | Latency (ms) |
|---|---|---|---|---|---|---|---|
| EDSR (teacher) ($n = 5$) | 27.385 ± 0.089 | 0.8293 ± 0.0023 | 0.1248 ± 0.0048 | 4.9292 | 1.9699 | 3.9398 | 0.895 |
| span_baseline ($n = 5$) | 26.950 ± 0.317 | 0.8178 ± 0.0102 | 0.1442 ± 0.0106 | 0.4263 | 0.1701 | 0.3402 | 0.566 |
| **span_tiny** ($n = 5$) | **27.055 ± 0.043** | **0.8196 ± 0.0012** | **0.1556 ± 0.0018** | **0.230** | **0.0918** | **0.1835** | **0.256** |
| span_large ($n = 5$) | 27.191 ± 0.034 | 0.8243 ± 0.0009 | 0.1515 ± 0.0015 | 0.417 | 0.1664 | 0.3328 | 0.464 |
| RLFN (Track A, unmodified, secondary) | 27.210 | 0.8245 | 0.1379 | 0.3278 | 0.1239 | 0.2479 | 0.684 |
| RLFN-adapted (Track A) | 27.192 | 0.8251 | 0.1335 | 0.3278 | 0.1241 | 0.2482 | 0.681 |
| ECBSR (Track A) | 26.074 | 0.7883 | 0.1722 | 0.0167 | 0.0067 | 0.0133 | 0.137 |
| SAFMN (Track A) | 27.355 | 0.8287 | 0.1350 | 0.2395 | 0.0940 | 0.1881 | 1.824 |
| SMFANet (Track A) | 20.906 | 0.6038 | 0.3104 | 0.0844 | 0.0303 | 0.0605 | 1.851 |
| RLFN-adapted (Track B, distilled) ($n = 5$) | 27.034 ± 0.061 | 0.8188 ± 0.0018 | 0.1539 ± 0.0015 | 0.3278 | 0.1241 | 0.2482 | 0.710 |
| ECBSR (Track B, distilled) ($n = 5$) | 25.957 ± 0.070 | 0.7846 ± 0.0021 | 0.1839 ± 0.0037 | 0.0167 | 0.0067 | 0.0133 | 0.135 |
| SAFMN (Track B, distilled) ($n = 5$) | 27.217 ± 0.043 | 0.8243 ± 0.0014 | 0.1516 ± 0.0025 | 0.2395 | 0.0940 | 0.1881 | 1.799 |
| SMFANet (Track B, distilled) ($n = 5$) | 26.330 ± 0.108 | 0.7967 ± 0.0038 | 0.1812 ± 0.0065 | 0.0844 | 0.0303 | 0.0605 | 1.878 |
| SAFMN half-depth (Track B, distilled) | 27.002 | 0.8176 | 0.1564 | 0.1281 | 0.0503 | 0.1007 | 1.103 |
| SMFANet half-depth (Track B, distilled) | 26.140 | 0.7900 | 0.1935 | 0.0477 | 0.0173 | 0.0347 | 1.125 |

Within the SPAN family itself, span_tiny's mean PSNR (27.055±0.043 dB) is nominally higher than span_baseline's (26.950±0.317 dB) despite roughly half the parameters – a gap smaller than it first appears once span_baseline's substantially larger seed-to-seed variance is accounted for (nearly 7× span_tiny's standard deviation), and plausibly related to span_baseline's fine-tuning from a fixed pretrained initialization (Section 3.2) into a $20 \times 20$-input regime well outside that pretraining's original design range, versus span_tiny's from-scratch training specifically for this resolution; this is offered as a plausible account of the direction, not a confirmed mechanism, and does not affect the compression-safety claim, which concerns downstream accuracy rather than PSNR. span_tiny is the second-fastest model, behind only ECBSR, and third-smallest by parameter count and by MACs/FLOPs alike, behind ECBSR and SMFANet on both hardware-independent measures. On PSNR, extending the four Track-B contextual architectures to the same $n = 5$ SR-training seeds as span_tiny (rather than the single run used in an earlier comparison) gives a closer picture than a point estimate alone suggested: span_tiny (27.055±0.043 dB) and RLFN-adapted (27.034±0.061 dB) are statistically indistinguishable point estimates, both behind only SAFMN (27.217±0.043 dB) and both clearly ahead of SMFANet

(26.330±0.108 dB) and ECBSR (25.957±0.070 dB). span_tiny's PSNR is therefore not top-ranked, but it is no longer accurate to call it low-ranked either; as Table 11 shows next, its downstream-accuracy standing (tied with the top two, significantly ahead of the bottom two) now broadly tracks this PSNR grouping rather than contradicting it, though the two measures are not identical – span_tiny's case rests on matching the top performers at a fraction of their parameter count and latency, not on outranking them on either metric. SMFANet's pixel-loss-only (Track A) result (20.906 dB, still $n = 1$) was investigated with a dedicated debugging script. The checkpoint itself was confirmed healthy, and the low score reflects SMFANet's difficulty converging under pure pixel supervision at 20×20 resolution. With distillation supervision (Track B), SMFANet recovers to 26.330 dB (+5.4 dB), and Track B results are therefore used for all comparative claims involving SMFANet. The half-depth SAFMN/SMFANet rows support the attention-parameterization ablation (Section 5.8).

**Table 11** span_tiny vs. four other Track-B lightweight SR architectures, downstream accuracy, EarVN1.0, all eleven recognition backbones, $n = 5$ seeds. $d$ is span_tiny minus the other architecture (positive = span_tiny ahead); $p_{\mathrm{Bonf}}$ corrected across the 6-comparison family reported by each backbone's own aggregation (Section 3.5). "(equiv.)" indicates TOST-confirmed equivalence within ±1 percentage point.

| Backbone | vs. RLFN-adapted | vs. SAFMN | vs. ECBSR | vs. SMFANet |
|---|---|---|---|---|
| EfficientNet-B0 | $d$=−0.79, n.s. | $d$=−0.89, n.s. | $d$=+3.37, **sig.** | $d$=+6.97, **sig.** |
| GhostNet-100 | $d$=−0.98, n.s. | $d$=+0.24, n.s. (tiny ahead, equiv.) | $d$=+2.94, **sig.** | $d$=+2.30, **sig.** |
| LCNet-100 | $d$=−0.15, n.s. (equiv.) | $d$=−0.39, n.s. | $d$=+2.72, **sig.** | $d$=−0.14, n.s. |
| MobileNetV2 | $d$=−0.17, n.s. | $d$=−0.24, n.s. (equiv.) | $d$=+3.79, **sig.** | $d$=+2.08, trend |
| MobileNetV3-Large | $d$=−0.14, n.s. (equiv.) | $d$=−0.69, n.s. | $d$=+0.73, n.s. | $d$=+2.56, **sig.** |
| MobileNetV3-Small | $d$=−0.22, n.s. | $d$=+0.11, n.s. (tiny ahead) | $d$=+2.41, **sig.** | $d$=+1.37, n.s. (tiny ahead) |
| MobileOne-S0 | $d$=+0.17, n.s. (tiny ahead) | $d$=+1.10, n.s. (tiny ahead) | $d$=+1.65, n.s. | $d$=+1.28, n.s. (tiny ahead) |
| RegNetY-400MF | $d$=+0.07, n.s. (tiny ahead, equiv.) | $d$=+0.53, n.s. (tiny ahead) | $d$=+4.39, **sig.** | $d$=+2.47, **sig.** |
| ResNet-18 | $d$=−0.56, n.s. | $d$=+0.18, n.s. (tiny ahead, equiv.) | $d$=+1.85, trend | $d$=+1.49, n.s. (tiny ahead) |
| ShuffleNetV2-1.0x | $d$=−0.18, n.s. | $d$=−0.02, n.s. (equiv.) | $d$=+2.97, **sig.** | $d$=+1.77, trend |
| SqueezeNet-1.1 | $d$=−0.11, n.s. (equiv.) | $d$=−0.69, n.s. | $d$=+2.61, **sig.** | $d$=+2.11, trend |

Across all eleven backbones, span_tiny is never significantly beaten by any of the four architectures compared. Against RLFN-adapted, span_tiny is statistically tied in 11/11 backbones, nominally ahead in 2/11 (MobileOne-S0, RegNetY-400MF), and TOST confirms genuine equivalence in 4/11 (LCNet-100, MobileNetV3-Large, RegNetY-400MF, SqueezeNet-1.1, $p_{\mathrm{TOST}}$ 0.0047–0.0382); the remaining 7/11 are tied but not TOST-confirmed, so that comparison stays an open, underpowered question for those backbones rather than a confirmed tie. Against SAFMN, span_tiny is likewise tied in 11/11, nominally ahead in 5/11 (GhostNet-100, MobileNetV3-Small, MobileOne-S0, RegNetY-400MF, ResNet-18), and TOST-confirmed equivalent in 4/11 (GhostNet-100, MobileNetV2, ResNet-18, ShuffleNetV2-1.0x, $p_{\mathrm{TOST}}$ 0.0025–0.0401). Against ECBSR, span_tiny wins with significance in 8/11 backbones, at trend level in a ninth (ResNet-18, $p_{\mathrm{Bonf}}$=0.0862), and is statistically tied in the remaining two (MobileNetV3-Large, MobileOne-S0) – the first backbones in this comparison, across either the original five or the six added later, where span_tiny does not clearly beat ECBSR. Against SMFANet, span_tiny wins with significance in 4/11, at trend level in 3/11 (MobileNetV2, ShuffleNetV2-1.0x, SqueezeNet-1.1), and is tied in the remaining 4/11, nominally ahead in three of those (MobileNetV3-Small, MobileOne-S0, ResNet-18) and nominally (non-significantly) behind in one (LCNet-100). This broadly tracks, rather than contradicts, the PSNR/SSIM picture in Table 10 once all five architectures are measured at the same $n = 5$ standard (span_tiny statistically indistinguishable from RLFN-adapted, both behind only SAFMN, both clearly ahead of ECBSR/SMFANet): on the metric the paper is actually organized around, span_tiny is competitive with the best-performing alternatives and significantly outperforms the two weakest ones in most, though no longer quite all, backbones. The stronger evidence in this paper that pixel-fidelity metrics are an imperfect proxy for recognition-relevant SR quality comes not from this cross-architecture ranking but from the qualitative, within-image analysis (Section 5.1), where a near-tied PSNR is shown to be driven by accurate reconstruction of hair and background rather than the ear region the recognition task actually depends on. This comparison remains contextual to the paper's central claim (Sections 5.1–5.2), which concerns depth compression *within* the SPAN family, not a claim that span_tiny is the best-performing lightweight SR architecture in absolute terms.

## 5.5 Auxiliary validation: real low-resolution holdout

On a held-out set of 173 genuinely low-resolution ear images (rather than synthetically downsampled ones), an initial screen on mobilenet_v2 alone found that downstream accuracy *reverses* the main result's ordering (no-SR scoring above both SR variants). Extending this check to all eleven recognition backbones (Table 12), at $n = 5$ seeds each, shows this reversal is *not* a general property of the real-low-resolution regime: it replicates on only some backbones and is absent, or itself reversed, on others.

**Table 12** Real low-resolution holdout, identity accuracy, EarVN1.0 (173 images), $n = 5$ seeds, all eleven recognition backbones. "vs. baseline" and "vs. tiny" report span_baseline−no-SR and span_tiny−no-SR (Cohen's $d$, $p_{\text{Bonf}}$, family of 3 comparisons per backbone); a negative $d$ means no-SR scores higher (the reversal).

| Backbone | no-SR | span_baseline | span_tiny | vs. baseline ($d$, $p_{\text{Bonf}}$) | vs. tiny ($d$, $p_{\text{Bonf}}$) |
|---|---|---|---|---|---|
| EfficientNet-B0 | 0.1514 | 0.1422 | 0.1572 | $d$=−0.33, $p$=1.0 (n.s., reversed) | $d$=0.16, $p$=1.0 (n.s.) |
| GhostNet-100 | 0.1145 | 0.1411 | 0.1283 | $d$=0.62, $p$=0.711 (n.s.) | $d$=0.48, $p$=1.0 (n.s.) |
| LCNet-100 | 0.0983 | 0.1387 | 0.1329 | $d$=1.44, $p$=0.096 (trend) | $d$=0.90, $p$=0.340 (n.s.) |
| MobileNetV2 | 0.1757 | 0.1480 | 0.1538 | $d$=−0.79, $p$=0.453 (n.s., reversed) | $d$=−0.96, $p$=0.296 (n.s., reversed) |
| MobileNetV3-Large | 0.1249 | 0.1503 | 0.1433 | $d$=0.79, $p$=0.461 (n.s.) | $d$=0.62, $p$=0.717 (n.s.) |
| MobileNetV3-Small | 0.0752 | 0.1226 | 0.1191 | $d$=1.62, $p$=0.067 (trend) | $d$=2.04, $p$=0.031 (**sig.**) |
| MobileOne-S0 | 0.1214 | 0.1341 | 0.1306 | $d$=0.54, $p$=0.888 (n.s.) | $d$=0.25, $p$=1.0 (n.s.) |
| RegNetY-400MF | 0.1480 | 0.1480 | 0.1457 | $d$=0.00, $p$=1.0 (n.s., tied) | $d$=−0.12, $p$=1.0 (n.s.) |
| ResNet-18 | 0.1376 | 0.0983 | 0.1040 | $d$=−1.84, $p$=0.044 (**sig., reversed**) | $d$=−1.40, $p$=0.106 (n.s., reversed) |
| ShuffleNetV2-1.0x | 0.1029 | 0.1271 | 0.1156 | $d$=0.72, $p$=0.544 (n.s.) | $d$=0.41, $p$=1.0 (n.s.) |
| SqueezeNet-1.1 | 0.0971 | 0.1040 | 0.0994 | $d$=0.32, $p$=1.0 (n.s.) | $d$=0.08, $p$=1.0 (n.s.) |

Only 3 of the 11 backbones show the reversal direction at all (EfficientNet-B0, MobileNetV2, ResNet-18), one is exactly tied (RegNetY-400MF), and the remaining 7 show the expected direction (SR still helps on real low-resolution images). Of these vs. baseline comparisons only one reaches this paper's significance threshold: the reversal itself, on ResNet-18 ($d$=−1.84, $p_{\text{Bonf}}$=0.044). Two more reach trend level in the *opposite* direction, SR nominally helping on real low-resolution images (MobileNetV3-Small, $d$=1.62, $p_{\text{Bonf}}$=0.067; LCNet-100, $d$=1.44, $p_{\text{Bonf}}$=0.096), suggestive but not confirmed at this paper's own threshold. This argues against the reversal being a general property of the real-low-resolution regime: all eleven backbones are evaluated on the exact same 173 images and the exact same span_baseline/span_tiny reconstructions of them, so if the effect were purely a property of the images or the SR output, every backbone would show the same direction. It does not. The reversal is instead backbone-dependent – some recognition architectures are more sensitive than others to whatever artifact the resolution mismatch introduces into the SR reconstruction, in the same spirit as the backbone-dependent SR-training-seed sensitivity documented in Section 3.6. `span_tiny` vs. `span_baseline` itself remains non-significant on every one of the eleven backbones (not tabulated separately), consistent with the paper's central claim that the specific cost of compression is near zero on this holdout too, independent of whether a given backbone shows the no-SR/SR reversal.

The likely mechanism behind the reversal, where it occurs, is still a resolution mismatch: every one of the 173 holdout images has a native short side strictly below 20 px (range 13–19 px, mean 17.8 px), below the resolution the SR models were trained to receive as input, and many additionally have non-square aspect ratios (mean long side 25.7 px against a 17.8 px short side) that cause the shared letterbox-resize routine (Section 3.2) to shrink the short side further still before the SR model sees the image. Under this distribution mismatch, a model trained to confidently reconstruct fine detail from one specific, clean synthetic degradation can hallucinate plausible-looking but incorrect high-frequency structure, while plain bicubic upsampling (the no-SR path) never attempts this reconstruction and therefore cannot introduce it – a known failure mode in the SR literature (models trained on synthetic degradation transferring poorly to real-world degradation). What the eleven-backbone result adds is that this hallucination risk is not uniformly harmful to every downstream classifier: a recognition backbone's own robustness to exactly this kind of high-frequency artifact evidently varies, so the same SR output can cost ResNet-18 accuracy while helping MobileNetV3-Small and LCNet-100.

No mechanism for this backbone-dependence is *confirmed* here, but it is not entirely unexplained either: among the eleven backbones, ResNet-18 is the only one with a statistically solid reversal on this holdout (the two other nominal reversals, EfficientNet-B0 and MobileNetV2, are both non-significant, Table 12), and ResNet-18 is independently the one backbone already flagged elsewhere in this paper (Section 5.6.2) as not fitting the compression-safety pattern cleanly on AWEx's frozen-backbone-transfer protocol – a completely different dataset and a completely different kind of distribution shift (cross-dataset transfer with a frozen backbone, rather than a resolution mismatch within EarVN1.0). The same backbone standing out under two unrelated distribution-shift stress tests is a real, data-grounded pattern, consistent with a general-fragility account: of the eleven backbones, ResNet-18 alone uses full (non-depthwise-separable) convolutions throughout and predates the mobile-efficiency-driven design choices (NAS search, depthwise-separable convolutions, squeeze-excite, structural reparameterization) common to the other ten, any of which could plausibly affect robustness to out-of-distribution input independent of raw accuracy. Two more specific candidate accounts were checked and did not hold up: architecture family does not cleanly separate the reversing backbones from the rest (two of the three nominal-reversal backbones – EfficientNet-B0, MobileNetV2 – do use depthwise-separable convolutions, the same family as several non-reversing backbones), and the magnitude of a backbone's own SR benefit on clean synthetic data (Table 4) does not predict reversal either (MobileNetV2 shows the largest such benefit of any backbone and reverses; LCNet-100 shows a comparably large benefit and instead shows a trend-level improvement on this holdout). The general-fragility account is therefore offered as the most consistent candidate explanation given the evidence gathered across this paper, not a confirmed mechanism, and both it and the resolution-mismatch account are discussed further as a boundary condition on the paper's central claim in Section 7 rather than treated as evidence against it, since the main claim concerns the synthetic bicubic-downsampling regime defined in Section 3 and this auxiliary check probes a related but distinct regime the main experiments were never designed to answer. Gender accuracy on this holdout, and the crossed-seed replication check described for the original mobilenet_v2 screen, have not yet been extended to the other ten backbones; both remain MobileNetV2-only observations pending further work.

### 5.5.1 Mitigation for the boundary condition: degradation-augmented training

Rather than leaving the resolution-mismatch boundary condition as an unaddressed limitation, a direct mitigation was implemented and tested: both span_tiny and span_baseline were retrained with the LR input generated by a randomized real-world-style degradation pipeline (Gaussian blur, additive noise, and JPEG compression, each resampled independently per training sample per epoch) rather than clean bicubic downsampling alone, following the classical degradation-modeling approach used in BSRGAN [56] rather than the more elaborate two-stage pipeline of Real-ESRGAN [57], to keep the change tractable as a single-variable ablation. Only the training-time LR generation changes; the validation split, the distillation recipe, and the frozen teacher are all unchanged from the main recipe, isolating the effect of degradation diversity specifically. Table 13 first checks that this retraining does not disturb the paper's central claim, on the original screening backbone (MobileNetV2) and four further backbones added to test generality across architecture families (ResNet-18, EfficientNet-B0, GhostNet-100, ShuffleNetV2-1.0x): the paired $t$-test finds no significant difference between span_tiny and span_baseline under the new recipe for four of the five backbones (MobileNetV2, GhostNet-100, ResNet-18, ShuffleNetV2-1.0x, all $p_{\mathrm{Bonf}} \geq 0.18$), and a trend-level cost for the fifth (EfficientNet-B0, $d = -1.15$, $p_{\mathrm{Bonf}} = 0.061$) – consistent with EfficientNet-B0 already being the one backbone that also shows a real cost under the original bicubic-only recipe (Table 6). TOST does not confirm equivalence within $\pm 1$ percentage point for any of the five ($p_{\mathrm{TOST}}$ ranging 0.097–0.329), consistent with the same four EarVN1.0 backbones in Table 4 under the original recipe, where equivalence is likewise not confirmed at $n = 5$. This sanity check is read the same way as that main comparison: no evidence of a cost under the new recipe for four of five backbones, a trend-level cost for the one backbone already flagged as an exception, short of a fully confirmed equivalence claim for any of them – at a modest cost in absolute accuracy on the clean synthetic benchmark relative to the original bicubic-only recipe (mobilenet_v2: 49.92% and 49.80% here vs. 51.25%

and 50.26% in Table 4), an expected accuracy/robustness trade-off from training on a harder, more varied degradation distribution.

**Table 13** Compression-safety sanity check under degradation-augmented training, EarVN1.0, $n$=5 seeds, five backbones. $p_{\mathrm{TOST}}$ is the equivalence test (±1 percentage point margin, Section 3.5).

| Backbone | span_baseline (degrad.-aug.) | span_tiny (degrad.-aug.) | $d$, $p_{\mathrm{Bonf}}$ | $p_{\mathrm{TOST}}$ |
|---|---|---|---|---|
| MobileNetV2 | 0.4992 ± 0.0114 | 0.4980 ± 0.0016 | $d$=−0.09, $p$=0.848 (n.s.) | 0.097 (not equiv.) |
| EfficientNet-B0 | 0.5560 ± 0.0071 | 0.5475 ± 0.0015 | $d$=−1.15, $p$=0.061 (trend) | 0.329 (not equiv.) |
| GhostNet-100 | 0.5141 ± 0.0063 | 0.5082 ± 0.0052 | $d$=−0.72, $p$=0.182 (n.s.) | 0.170 (not equiv.) |
| ResNet-18 | 0.4863 ± 0.0122 | 0.4833 ± 0.0073 | $d$=−0.28, $p$=0.562 (n.s.) | 0.111 (not equiv.) |
| ShuffleNetV2-1.0x | 0.4851 ± 0.0072 | 0.4808 ± 0.0106 | $d$=−0.40, $p$=0.418 (n.s.) | 0.146 (not equiv.) |

Table 14 re-evaluates both retrained models on the same 173-image real-low-resolution holdout, at $n$=5 seeds.

**Table 14** Real low-resolution holdout under degradation-augmented training vs. the original bicubic-only recipe (Table 12), EarVN1.0 (mobilenet_v2), $n$=5 seeds. Gender accuracy uses EarVN1.0's folder-range gender convention (Section 4.1), not an independently verified attribute.

| Condition | Identity (bicubic-only) | Identity (degrad.-aug.) | Gender (bicubic-only) | Gender (degrad.-aug.) |
|---|---|---|---|---|
| no-SR | 0.1757 ± 0.0194 | – | 0.8451 ± 0.0103 | – |
| span_baseline | 0.1480 ± 0.0194 | 0.1318 ± 0.0193 | 0.8301 ± 0.0113 | 0.8127 ± 0.0145 |
| span_tiny | 0.1538 ± 0.0264 | 0.1572 ± 0.0384 | 0.8312 ± 0.0155 | 0.8509 ± 0.0229 |

**Table 15** Real low-resolution holdout under degradation-augmented training, pairwise comparisons ($d$, $p_{\mathrm{Bonf}}$; family of 3 per metric), $n$=5 seeds. Gender accuracy uses EarVN1.0's folder-range gender convention (Section 4.1), not an independently verified attribute.

| Comparison | Identity accuracy | Gender accuracy |
|---|---|---|
| span_baseline − no-SR | $d$=−1.41, $p$=0.104 (n.s.) | $d$=−3.08, $p$=0.007 (sig.) |
| span_tiny − no-SR | $d$=−0.35, $p$=1.0 (n.s., gap narrowed from $d$=−0.96) | $d$=0.23, $p$=1.0 (n.s., span_tiny nominally ahead) |
| span_tiny − span_baseline | $d$=0.82, $p$=0.421 (n.s.) | $d$=1.45, $p$=0.095 (trend, span_tiny ahead) |

The effect of degradation-augmented training is asymmetric between the two architectures. For span_baseline, the mitigation does not help and, on gender accuracy, makes the gap to no-SR significantly worse than under the original recipe ($d$=−3.08, $p_{\mathrm{Bonf}}$=0.007, compared to $d$=−1.43 under the original bicubic-only recipe on this same mobilenet_v2 screen, Section 5.5; gender accuracy on this holdout was a mobilenet_v2-only measurement at this stage, extended to four further backbones below). For span_tiny, the mitigation shows a consistent, if not fully significant, improvement on both metrics: on identity accuracy, the gap to no-SR narrows from $d$=−0.96 to $d$=−0.35; on gender accuracy, span_tiny under degradation-augmented training is nominally *ahead* of no-SR (0.8509 vs. 0.8451, $d$=0.23), a reversal of the original direction, and now ahead of span_baseline under the same recipe at trend level ($d$=1.45, $p_{\mathrm{Bonf}}$=0.095), a comparison that was flat ($d$=0.04) before this mitigation. None of the individual comparisons reaches significance against no-SR after

Bonferroni correction at $n$=5, so this is reported as a partial, directionally consistent mitigation specific to the compressed model, not a resolved boundary condition: degradation-augmented training measurably helps span_tiny's robustness to genuinely low-resolution real images without disturbing the compression-safety claim, but does not by itself close the gap to no-SR with statistical certainty, and does not transfer the same benefit to span_baseline under the specific degradation recipe tested here.

A candidate mechanism for the asymmetry itself (that span_baseline's larger capacity permits more confident, and potentially more incorrect, high-frequency reconstruction under distribution shift than span_tiny's) was tested directly with a paired Laplacian-variance analysis across all 173 holdout images, using the gain in sharpness relative to no-SR as a proxy for the amount of high-frequency content each model adds; this proxy quantifies added detail but, absent a clean HR reference for these genuinely low-resolution images, cannot on its own distinguish correct detail recovery from hallucination. The paired effect size between span_baseline and span_tiny is negligible under both the original recipe ($d$=0.06) and degradation-augmented training ($d$=−0.07), and across the four conditions, mean sharpness gain correlates positively rather than negatively with mean identity accuracy (Spearman $\rho$=0.20, $n$=4), the opposite of what the capacity/hallucination account predicts. This candidate mechanism therefore finds no support, and the cause of the span_tiny/span_baseline asymmetry under degradation-augmented training is left open rather than assigned an unverified explanation.

The mitigation's effectiveness itself turns out to be backbone-dependent when the same holdout evaluation is extended beyond MobileNetV2 to the same four further backbones as the depth sweep (Section 5.2.1). Table 16 reports condition means and Table 17 the pairwise comparisons.

**Table 16** Real low-resolution holdout under degradation-augmented training, four further backbones (identity and gender accuracy, mean ± std. dev., $n$=5 seeds each; original MobileNetV2 results in Table 14). Gender accuracy uses EarVN1.0's folder-range gender convention (Section 4.1), not an independently verified attribute.

| Backbone | no-SR (id.) | span_baseline (id.) | span_tiny (id.) | no-SR (gender) | span_baseline (gender) | span_tiny (gender) |
|---|---|---|---|---|---|---|
| EfficientNet-B0 | 0.1514 ± 0.0138 | 0.1468 ± 0.0389 | 0.1491 ± 0.0325 | 0.8139 ± 0.0150 | 0.8312 ± 0.0180 | 0.8035 ± 0.0274 |
| GhostNet-100 | 0.1145 ± 0.0138 | 0.1514 ± 0.0150 | 0.1376 ± 0.0063 | 0.7965 ± 0.0165 | 0.8254 ± 0.0175 | 0.8231 ± 0.0088 |
| ResNet-18 | 0.1376 ± 0.0198 | 0.0983 ± 0.0108 | 0.1064 ± 0.0181 | 0.7861 ± 0.0142 | 0.7352 ± 0.0243 | 0.7572 ± 0.0303 |
| ShuffleNetV2-1.0x | 0.1029 ± 0.0214 | 0.1237 ± 0.0167 | 0.1410 ± 0.0264 | 0.7572 ± 0.0268 | 0.8197 ± 0.0328 | 0.8081 ± 0.0263 |

**Table 17** Real low-resolution holdout under degradation-augmented training, pairwise comparisons for four further backbones ($d$, family of 3 comparisons per metric per backbone, $n$=5 seeds; original MobileNetV2 results in Table 15). Gender accuracy uses EarVN1.0's folder-range gender convention (Section 4.1), not an independently verified attribute.

| | Identity accuracy | | | Gender accuracy | | |
|---|---|---|---|---|---|---|
| Backbone | baseline−no-SR | tiny−no-SR | tiny−baseline | baseline−no-SR | tiny−no-SR | tiny−baseline |
| EfficientNet-B0 | $d$=−0.10, n.s. | $d$=−0.08, n.s. | $d$=0.04, n.s. | $d$=0.87, n.s. | $d$=−0.30, n.s. | $d$=−0.64, n.s. |
| GhostNet-100 | $d$=1.99, **sig.** | $d$=2.52, **sig.** | $d$=−0.79, n.s. | $d$=1.27, n.s. | $d$=1.84, **sig.** | $d$=−0.17, n.s. |
| ResNet-18 | $d$=−2.45, **sig.** | $d$=−1.00, n.s. | $d$=0.30, n.s. | $d$=−2.37, **sig.** | $d$=−0.78, n.s. | $d$=0.68, n.s. |
| ShuffleNetV2-1.0x | $d$=1.99, **sig.** | $d$=2.74, **sig.** | $d$=0.82, n.s. | $d$=1.42, n.s. | $d$=1.45, trend | $d$=−0.28, n.s. |

Two of the four further backbones show a pattern the original MobileNetV2 hold-out did not: no reversal at all under the degradation-augmented recipe, with SR significantly *helping* relative to no-SR on identity accuracy for both span_baseline and span_tiny (GhostNet-100: $d$=1.99 and 2.52; ShuffleNetV2-1.0x: $d$=1.99 and 2.74; all $p_{\mathrm{Bonf}}<0.05$), with gender accuracy pointing the same direction on both (significant for GhostNet-100's span_tiny comparison, trend-level for ShuffleNetV2-1.0x's). EfficientNet-B0 shows no significant effect in either direction on either metric – a flat result. ResNet-18 is the one backbone where the mitigation clearly does not help: span_baseline remains significantly worse than no-SR on both identity ($d$=−2.45, $p_{\mathrm{Bonf}}$=0.0163) and gender ($d$=−2.37, $p_{\mathrm{Bonf}}$=0.0182) accuracy even under the degradation-augmented recipe, with span_tiny nominally, but not significantly, closer to no-SR on both metrics – the same directionally-consistent-but-unconfirmed pattern already reported for MobileNetV2 above. This is not an isolated result: ResNet-18 is the same backbone identified in Section 5.5 as the only one with a statistically solid reversal under the *original*, non-augmented recipe (Table 12), and the same backbone independently flagged elsewhere in this paper (Section 5.6.2) as not fitting the compression-safety pattern cleanly on AWEx's frozen-backbone-transfer protocol. A mitigation designed and validated on one screening backbone does not automatically transfer to the specific backbone independently identified as most fragile to this class of distribution shift, which is itself informative: whatever makes ResNet-18 more sensitive to the resolution mismatch is not remedied by exposing training to more diverse synthetic degradation alone, at least not under the specific recipe tested here.

## 5.6 Cross-dataset generalization (AWEx)

All AWEx results below were generated after every correction described in Section 3.7 and every statistical-methodology refinement described in Section 3.5 had already been incorporated; no AWEx figure from an earlier, uncorrected stage of this study is reused anywhere in this paper.

### 5.6.1 From-scratch track

Table 18 reports results from models trained from scratch on AWEx (no transfer of weights or teacher checkpoints from EarVN1.0), across all eleven recognition backbones. $p_{\mathrm{Bonf}}$ is corrected across the 3-comparison family per backbone (lr–baseline, lr–tiny, baseline–tiny), matching Table 4's convention.

An earlier stage of this study, predating the corrections in Section 3.7, had reported ResNet-18's from-scratch baseline scoring *below* the no-SR condition on this track, the only off-trend result across both datasets at that stage. This does not replicate under the corrected setup (Table 18: baseline > lr for ResNet-18, +0.6 percentage points, though not significant at $n = 5$); the earlier result is attributed to the since-corrected issues rather than to a property of the backbone, and is not carried forward as a limitation. `span_tiny` vs. `span_large` on this track shows no statistically significant difference in 11/11 backbones (Section 5.2), further corroborating the depth-compression finding on a second, independent dataset. Critically, `tiny vs.`

**Table 18** Downstream identity accuracy on AWEx, from-scratch training, $n = 5$ seeds, all eleven recognition backbones. $p_{\mathrm{TOST}}$ is the equivalence test for tiny vs. baseline (±1 percentage point margin, Section 3.5); "equiv." requires $p_{\mathrm{TOST}} < 0.05$, a distinct claim from "n.s."

| Backbone | lr | span_baseline | span_tiny | baseline>lr | tiny>lr | tiny vs. baseline | $p_{\mathrm{TOST}}$ |
|---|---|---|---|---|---|---|---|
| EfficientNet-B0 | 0.1290 | 0.1619 | 0.1524 | sig. ($p$=0.0011) | sig. ($p$=0.0127) | n.s. ($p$=0.1266) | 0.435, not equiv. |
| GhostNet-100 | 0.0892 | 0.1234 | 0.1304 | sig. ($p$=0.0129) | sig. ($p$=0.0014) | n.s. ($p$=0.2226, tiny higher) | 0.183, not equiv. |
| LCNet-100 | 0.0867 | 0.1056 | 0.1094 | n.s. ($p$=0.3114) | n.s. ($p$=0.1119) | n.s. ($p$=0.9184, tiny higher) | 0.067, not equiv. |
| MobileNetV2 | 0.0885 | 0.1196 | 0.1210 | sig. ($p$=0.0293) | trend ($p$=0.0543) | n.s. ($p$=1.0, tiny higher) | **0.024, equiv.** |
| MobileNetV3-Large | 0.0804 | 0.1157 | 0.1196 | sig. ($p$=0.0203) | sig. ($p$=0.0145) | n.s. ($p$=1.0, tiny higher) | 0.300, not equiv. |
| MobileNetV3-Small | 0.0888 | 0.1115 | 0.1084 | sig. ($p$=0.0208) | sig. ($p$=0.0116) | n.s. ($p$=0.698) | **0.019, equiv.** |
| MobileOne-S0 | 0.1462 | 0.1559 | 0.1615 | n.s. ($p$=0.118) | sig. ($p$=0.0003) | n.s. ($p$=0.2332, tiny higher) | 0.069, not equiv. |
| RegNetY-400MF | 0.1133 | 0.1511 | 0.1448 | sig. ($p$=0.0082) | trend ($p$=0.0739) | n.s. ($p$=1.0) | 0.302, not equiv. |
| ResNet-18 | 0.1748 | 0.1808 | 0.1860 | n.s. ($p$=0.6719) | n.s. ($p$=0.543) | n.s. ($p$=1.0, tiny higher) | 0.280, not equiv. |
| ShuffleNetV2-1.0x | 0.1091 | 0.1115 | 0.1210 | n.s. ($p$=1.0) | n.s. ($p$=0.3894) | n.s. ($p$=0.1489, tiny higher) | 0.441, not equiv. |
| SqueezeNet-1.1 | 0.1734 | 0.1689 | 0.1664 | n.s. ($p$=0.8355) | n.s. ($p$=0.1647) | n.s. ($p$=1.0) | 0.059, not equiv. |

`baseline` is non-significant in 11/11 backbones here too. Equivalence testing distinguishes the same two categories seen on EarVN1.0, now on a broader set: of the 11 non-significant `tiny vs. baseline` comparisons, 2 (MobileNetV2, MobileNetV3-Small, $p_{\mathrm{TOST}}$ 0.019–0.024) are TOST-confirmed equivalent within ±1 percentage point, while the remaining 9 are non-significant without being confirmed equivalent. `baseline>lr` reaches significance in 7/11 backbones and trend level in one more (RegNetY-400MF); `tiny>lr` reaches significance in 5/11 and trend level in one more (MobileNetV2).

An HR-ceiling check identical in protocol to Section 5.1.1 was also run on AWEx (from-scratch track, $n = 5$ seeds, all eleven backbones); Table 19 reports it, with the same 6-comparison Bonferroni family used there.

**Table 19** HR ceiling vs. the three reconstructed domains, downstream identity accuracy, AWEx, from-scratch training, $n = 5$ seeds, all eleven recognition backbones. $p_{\mathrm{Bonf}}$ corrected across the same 6-comparison family as Table 5.

| Backbone | hr (ceiling) | lr | span_baseline | span_tiny | hr vs. lr | hr vs. tiny |
|---|---|---|---|---|---|---|
| EfficientNet-B0 | 0.1675 | 0.1290 | 0.1619 | 0.1524 | sig. ($p$=0.0022) | n.s. ($p$=0.2038) |
| GhostNet-100 | 0.1434 | 0.0892 | 0.1234 | 0.1304 | sig. ($p$=0.0303) | n.s. ($p$=1.0) |
| LCNet-100 | 0.1238 | 0.0867 | 0.1056 | 0.1094 | trend ($p$=0.0801) | n.s. ($p$=0.1261) |
| MobileNetV2 | 0.1402 | 0.0885 | 0.1196 | 0.1210 | sig. ($p$=0.0111) | sig. ($p$=0.0019) |
| MobileNetV3-Large | 0.1315 | 0.0804 | 0.1157 | 0.1196 | sig. ($p$=0.0067) | n.s. ($p$=1.0) |
| MobileNetV3-Small* | 0.0951 | 0.0888 | 0.1115 | 0.1084 | n.s. ($p$=1.0) | trend ($p$=0.0911, tiny *higher*) |
| MobileOne-S0 | 0.1699 | 0.1462 | 0.1559 | 0.1615 | sig. ($p$=0.0138) | n.s. ($p$=0.2401) |
| RegNetY-400MF | 0.1430 | 0.1133 | 0.1511 | 0.1448 | n.s. ($p$=0.1673) | n.s. ($p$=1.0, tiny higher) |
| ResNet-18 | 0.2304 | 0.1748 | 0.1808 | 0.1860 | n.s. ($p$=0.1183) | n.s. ($p$=0.2512) |
| ShuffleNetV2-1.0x | 0.1371 | 0.1091 | 0.1115 | 0.1210 | sig. ($p$=0.0097) | n.s. ($p$=0.2231) |
| SqueezeNet-1.1 | 0.2094 | 0.1734 | 0.1689 | 0.1664 | sig. ($p$=0.0098) | sig. ($p$=0.0030) |

*For MobileNetV3-Small, `span_tiny` is nominally *above* the hr ceiling at trend level ($d$=+1.82, $p_{\mathrm{Bonf}}$=0.0911, not reaching this paper's significance threshold) – investigated directly (confusion matrices, training logs) rather than assumed to be noise, and traced to a validation-to-test generalization gap roughly five times larger for `hr` than for `span_tiny` on this backbone, not a computation error or evidence that SR carries more identity information than the true source; discussed further below.

The pattern here is far weaker and noisier than on EarVN1.0, consistent with AWEx's already-documented small per-subject sample size (Section 6). `hr>lr` reaches significance in 7 of 11 backbones, with one more at trend level (LCNet-100, $p_{\mathrm{Bonf}}$=0.0801); the remaining three are non-significant: MobileNetV3-Small, and, notably, RegNetY-400MF and ResNet-18, both of which show a nominally large hr-lr gap (5.6–5.7 points) that does not reach significance at $n = 5$, a power limitation rather than evidence the ceiling does not exist for these backbones. `hr` vs. `span_tiny` reaches significance in only 2 of 11 (MobileNetV2, SqueezeNet-1.1), both behaving as a ceiling should, with one more at trend level in the *opposite* direction: for MobileNetV3-Small, `span_tiny` is nominally *above* the hr ceiling at trend level ($d$=+1.82, $p_{\mathrm{Bonf}}$=0.0911, not reaching this paper's significance threshold), and `span_baseline` is nominally above it too, at an even weaker level ($d$=+1.69, $p_{\mathrm{Bonf}}$=0.1172). This trend-level reversal was investigated directly rather than left unexamined: recomputing identity accuracy from each seed's confusion matrix matches the reported values exactly (ruling out a computation or file-mixup error), and training logs show `hr`'s best-validation checkpoint reaches a mean validation accuracy (0.123) at least as high as `span_tiny`'s (0.114) but generalizes to the held-out test set substantially worse: the validation-to-test gap is roughly five times larger for `hr` (mean +2.8 percentage points, positive in 5 of 5 seeds) than for `span_tiny` (mean +0.6 points, positive in 4 of 5, reversing sign on the fifth). The most consistent account is that this backbone's identity-classification head, trained directly on `hr`'s uncompressed images under AWEx's very small per-subject sample (≈6.7 images/subject), overfits its own validation split more than it does when trained on `span_tiny`'s smoother, SR-reconstructed images – a generalization-gap effect specific to the classification head, not evidence that a super-resolved image carries more identity information than the true high-resolution source, which would have no plausible mechanism. SqueezeNet-1.1's ceiling comparisons being significant while its `lr` vs. `span_baseline`/`span_tiny` comparisons are not (Table 18) is consistent with the same backbone-specific pattern noted there. This AWEx check is reported as a weaker-power replication of the EarVN1.0 finding, not a contradiction of it: of the seven backbones reaching significance on any ceiling comparison here, all seven agree with the expected direction (`hr` above the reconstructed domains); the one apparent reversal (MobileNetV3-Small) does not reach this paper's own significance threshold, only a trend, and is reported at that strength on the dataset already flagged throughout this paper as underpowered.

Table 20 reports SR image quality for the from-scratch track.

**Table 20** SR image quality on AWEx, from-scratch training. span_baseline/span_tiny/span_large report mean±std across the same $n = 5$ SR-training seeds (42, 123, 2024, 44, 999) used for the SR-seed-variance analysis (Section 3.6); EDSR remains $n = 1$ (Section 7).

| Model | PSNR (dB) | SSIM | LPIPS↓ |
|---|---|---|---|
| EDSR (teacher) | 25.584 | 0.7687 | 0.1642 |
| span_baseline (span_official) ($n = 5$) | 25.241 ± 0.125 | 0.7594 ± 0.0052 | 0.1807 ± 0.0059 |
| **span_tiny** ($n = 5$) | **24.782 ± 0.104** | **0.7421 ± 0.0034** | **0.2345 ± 0.0062** |
| span_large ($n = 5$) | 24.513 ± 0.218 | 0.7305 ± 0.0077 | 0.2709 ± 0.0205 |

All four models score within a coherent 24.5–25.6 dB band with no anomaly, in contrast to an earlier stage of this study in which `span_large` had scored anomalously low (≈19 dB, 5–6 dB below every other architecture) with no corresponding downstream-accuracy effect; that anomaly is not present under the current measurements and is not carried forward as an open item. The contextual comparison against RLFN/ECBSR/SAFMN/SMFANet (Table 10, Table 11) was run on EarVN1.0 only in this phase of the study; extending it to AWEx is noted as a direction for future work (Section 7).

### 5.6.2 Transfer-learning tracks (primary result for AWEx)

Given AWEx's smaller per-subject sample size, two transfer-learning protocols initialized from EarVN1.0-trained weights are evaluated: full fine-tuning of the entire recognition backbone (Table 21), and a frozen-backbone variant that only fine-tunes the classification head (Table 22). Both are reported as primary results, because they answer a related but distinct question and, as discussed below, do not agree in every respect.

**Table 21** Downstream identity accuracy on AWEx, transfer-learning track (full fine-tuning), $n = 5$ seeds, all eleven recognition backbones. $p_{\text{Bonf}}$ corrected across the 3-comparison family per backbone (lr–baseline, lr–tiny, baseline–tiny), matching Table 4's convention; only `tiny vs. baseline`, the comparison central to this paper's claim, is tabulated, with `baseline`>`lr` and `tiny`>`lr` discussed in the main text. $p_{\text{TOST}}$ is the equivalence test ($\pm 1$ percentage point margin, Section 3.5).

| Backbone | lr | span_baseline (transfer) | span_tiny (transfer) | tiny vs. baseline | $p_{\text{TOST}}$ |
|---|---|---|---|---|---|
| EfficientNet-B0 | 0.1780 | 0.2063 | 0.1864 | n.s. ($p$=0.1613) | 0.875, not equiv. |
| GhostNet-100 | 0.1301 | 0.1643 | 0.1475 | n.s. ($p$=0.1207) | 0.854, not equiv. |
| LCNet-100 | 0.1308 | 0.1601 | 0.1528 | n.s. ($p$=0.9953) | 0.355, not equiv. |
| MobileNetV2 | 0.1178 | 0.1510 | 0.1402 | n.s. ($p$=0.625) | 0.543, not equiv. |
| MobileNetV3-Large | 0.1549 | 0.1762 | 0.1734 | n.s. ($p$=1.0) | 0.223, not equiv. |
| MobileNetV3-Small | 0.1091 | 0.1416 | 0.1469 | n.s. ($p$=1.0, tiny higher) | 0.223, not equiv. |
| MobileOne-S0 | 0.1234 | 0.1441 | 0.1364 | n.s. ($p$=0.8895) | 0.369, not equiv. |
| RegNetY-400MF | 0.0944 | 0.1147 | 0.1122 | n.s. ($p$=1.0) | 0.053, not equiv. |
| ResNet-18 | 0.1021 | 0.1196 | 0.1070 | n.s. ($p$=0.549) | 0.620, not equiv. |
| ShuffleNetV2-1.0x | 0.1220 | 0.1381 | 0.1402 | n.s. ($p$=1.0, tiny higher) | 0.138, not equiv. |
| SqueezeNet-1.1 | 0.1238 | 0.1353 | 0.1329 | n.s. ($p$=1.0) | 0.095, not equiv. |

Under full fine-tuning, `span_baseline` > `lr` reaches significance in 8/11 backbones and trend level in a ninth (MobileNetV3-Large), and `span_tiny` > `lr` in 3/11 (MobileNetV2, MobileNetV3-Small, RegNetY-400MF; $d$=3.22, 3.23, 2.80). Critically for this paper's central claim, `span_tiny` vs. `span_baseline` is non-significant in 11/11 backbones. TOST does not confirm genuine equivalence for any of the eleven backbones under this protocol ($p_{\text{TOST}}$ 0.053–0.905), a weaker showing than the from-scratch track's 2/11 confirmed. The compression-safety finding replicates under this protocol at the level of "not detected as significant" across all eleven backbones, though genuine equivalence is not confirmed for any of them here.

Under the frozen-backbone protocol, `span_baseline` > `lr` reaches significance in 6/11 backbones (LCNet-100, MobileNetV2, MobileNetV3-Large, MobileNetV3-Small, RegNetY-400MF, SqueezeNet-1.1) and trend level in two more (EfficientNet-B0, $p_{\text{Bonf}}$=0.0973; ShuffleNetV2-1.0x, $p_{\text{Bonf}}$=0.0604); `span_tiny` > `lr` reaches significance in 5/11 (EfficientNet-B0, GhostNet-100, LCNet-100, MobileNetV3-Large, MobileNetV3-Small). As with full fine-tuning, `span_tiny` vs. `span_baseline` reaches this paper's significance threshold in 0/11 backbones, with one at trend level

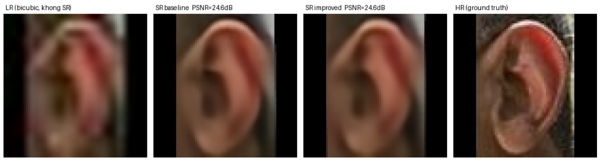
**Table 22** Downstream identity accuracy on AWEx, transfer-learning track (frozen backbone), $n = 5$ seeds, all eleven recognition backbones. $p_{\text{Bonf}}$ corrected across the 3-comparison family per backbone (lr–baseline, lr–tiny, baseline–tiny), matching Table 4's convention; only `tiny vs. baseline`, the comparison central to this paper's claim, is tabulated, with `baseline`>`lr` and `tiny`>`lr` discussed in the main text. $p_{\text{TOST}}$ is the equivalence test (±1 percentage point margin, Section 3.5).

| Backbone | lr | span_baseline (transfer) | span_tiny (transfer) | tiny vs. baseline | $p_{\text{TOST}}$ |
|---|---|---|---|---|---|
| EfficientNet-B0 | 0.1706 | 0.2045 | 0.2126 | n.s. ($p$=1.0, tiny higher) | 0.451, not equiv. |
| GhostNet-100 | 0.1367 | 0.1479 | 0.1647 | n.s. ($p$=0.871, tiny higher) | 0.676, not equiv. |
| LCNet-100 | 0.1385 | 0.1881 | 0.1846 | n.s. ($p$=1.0) | 0.242, not equiv. |
| MobileNetV2 | 0.1329 | 0.1609 | 0.1329 | n.s. ($p$=0.212) | 0.905, not equiv. |
| MobileNetV3-Large | 0.1710 | 0.2185 | 0.2161 | n.s. ($p$=1.0) | 0.288, not equiv. |
| MobileNetV3-Small | 0.1052 | 0.1444 | 0.1469 | n.s. ($p$=1.0, tiny higher) | 0.178, not equiv. |
| MobileOne-S0 | 0.1559 | 0.1885 | 0.1846 | n.s. ($p$=1.0) | 0.099, not equiv. |
| RegNetY-400MF | 0.0940 | 0.1703 | 0.1678 | n.s. ($p$=1.0) | 0.240, not equiv. |
| ResNet-18 | 0.1070 | 0.1154 | 0.0951 | n.s. ($p$=0.173) | 0.874, not equiv. |
| ShuffleNetV2-1.0x | 0.1241 | 0.1598 | 0.1654 | n.s. ($p$=1.0, tiny higher) | 0.348, not equiv. |
| SqueezeNet-1.1 | 0.1427 | 0.1790 | 0.1594 | trend ($p$=0.0994) | 0.904, not equiv. |

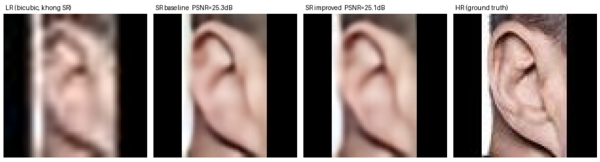
(SqueezeNet-1.1, $p_{\text{Bonf}}$=0.0994); TOST does not confirm equivalence in any of the 11, including SqueezeNet-1.1 itself (TOST result 0.904, far from confirming equivalence despite its trend-level NHST result). The compression-safety finding is therefore confirmed, at the "not detected as significant" strength, under both transfer protocols and the from-scratch track alike: 0/11 backbones show a significant accuracy cost from compression under any of the three protocols tested on AWEx; genuine equivalence is TOST-confirmed for 2 of the 33 backbone×protocol combinations overall (2 on the from-scratch track, 0 under full-fine-tune transfer, 0 under frozen-backbone transfer). The two protocols do disagree on secondary detail: under frozen backbone, `span_tiny` is nominally *above* `span_baseline` in 4/11 backbones (never significantly), MobileNetV2 shows tiny exactly tied with `lr`, and ResNet-18 is the one case across all AWEx protocols where `span_tiny` falls nominally below `lr` (not significant, $p_{\text{Bonf}}$=1.0). ResNet-18 remains, across all three AWEx protocols and now across all eleven backbones, the one case where `span_tiny` falls nominally below `lr` under any protocol; this is noted as a backbone-specific limitation (Section 7), not as evidence against the compression-safety claim, which does not depend on this secondary comparison.

An identical real-low-resolution-holdout check to Section 5.1 was also run on AWEx (mobilenet_v2, frozen-backbone checkpoints, $n = 1$): no-SR = 0.0, span_baseline= 0.0625, span_tiny= 0.0938. Unlike the EarVN1.0 holdout (Section 5.1), both SR variants clearly beat no-SR here rather than reversing it; span_tiny is nominally ahead of span_baseline, echoing the same-direction flip seen for this backbone-adjacent pattern elsewhere in this study (Tables 4, 22). With $n = 1$ and only 32 images (Section 7), this is not treated as contradicting the EarVN1.0 finding, only as insufficient evidence on its own either way.

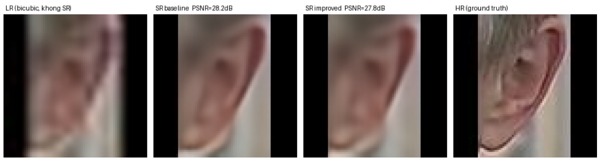
Figure 4 shows qualitative examples on AWEx test images, from the transfer-learning track (full fine-tuning protocol, Table 21), using the SR checkpoints fine-tuned specifically for that protocol. Across the 15 exported AWEx test samples, the per-image PSNR gap (span_baseline − span_tiny) ranges from −0.042 dB (sample index 13, span_tiny fractionally ahead) to +1.358 dB (sample index 8, the largest



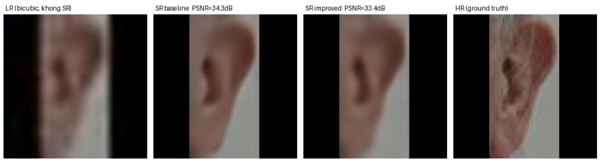

(a) baseline 24.602 dB/SSIM 0.7787, span_tiny 24.644 dB/SSIM 0.7733: span_tiny marginally ahead, minimum gap observed



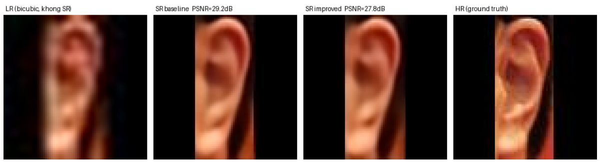

(b) baseline 25.318 dB/SSIM 0.8643, span_tiny 25.092 dB/SSIM 0.8557: gap 0.226 dB (Q1)




(c) baseline 28.190 dB/SSIM 0.8278, span_tiny 27.827 dB/SSIM 0.8168: gap 0.363 dB (median)




(d) baseline 34.258 dB/SSIM 0.9302, span_tiny 33.351 dB/SSIM 0.9218: gap 0.907 dB (Q3)




(e) baseline 29.204 dB/SSIM 0.8962, span_tiny 27.846 dB/SSIM 0.8745: largest gap observed across all 15 exported AWEx samples

**Fig. 4** Qualitative comparison on five AWEx test images, each showing (left to right) the bicubic-upsampled LR input (no SR), span_baseline output, span_tiny output, and the HR ground truth. Samples (a)–(e) are selected by PSNR-gap quartile (minimum, Q1, median, Q3, maximum) across the 15-sample export, from span_tiny marginally ahead (a) to the largest gap in favor of span_baseline (e).

gap observed), a narrower range than in an earlier export tied to a different (frozen-backbone) checkpoint, consistent with this being a genuinely different SR checkpoint rather than a re-labeling of the same one. The five panels shown are selected by PSNR-gap quartile across the full 15-sample export (minimum, Q1, median, Q3, maximum) rather than by visual inspection, so the selection is not biased toward either model; none of the five is grayscale.

The paper's central compression-safety claim (`span_tiny` statistically indistinguishable from `span_baseline`) is confirmed on a second, independent dataset under *three* separate protocols (from-scratch, full-fine-tune transfer, frozen-backbone transfer), with 0/11 backbones showing a significant cost under any of them; equivalence testing (TOST) confirms this is genuinely negligible for 2 of the 33 backbone×protocol combinations (MobileNetV2 and MobileNetV3-Small, both on the from-scratch track), with the remainder reported as not detected as significant rather than confirmed safe. The secondary question, whether SR helps over no-SR at all, is directionally positive across most backbone×protocol combinations but noisier and protocol-dependent in exactly which backbones reach significance, consistent with AWEx's smaller per-subject training sample (≈6.7 images/subject vs. substantially more on EarVN1.0) rather than a contradiction of the EarVN1.0 findings; this distinction is discussed further in Section 6. ResNet-18 is the one backbone that does not fit this pattern cleanly on AWEx (Section 5.6.2), a limitation discussed further in Section 7.

## 5.7 Additional mechanisms tested for further improvement: a negative-result ablation study

None of the three candidate mechanisms for improving span_tiny's downstream accuracy further – a richer distillation loss, a saliency-weighted identity-critical loss, and learned/differentiable block pruning as an alternative to span_tiny's fixed, hand-picked block selection – improves on the main recipe. Each was screened at reduced scale first, then validated to the same statistical standard as the main result ($n = 5$ seeds, all eleven recognition backbones, EarVN1.0).

### 5.7.1 Feature-level distillation ("KD v2")

This mechanism adds a feature-level distillation term (matching intermediate SPAN features between student and teacher) to the existing pixel+distillation recipe. A related variant additionally combining this with a multi-judge identity loss (an ensemble of independently-trained recognition backbones providing identity supervision) was screened alongside it but performed no better during screening; the feature-only variant, the stronger candidate, is the one carried forward to the full $n = 5$-seed validation reported here. Table 25 (Supplementary Material) compares it against the main recipe (span_tiny).

No backbone shows a statistically significant difference in either direction ($p_{\mathrm{Bonf}}$ 0.35–1.0), and the sign of the effect is inconsistent across backbones (positive in 5/11, negative in 6/11). The effect sizes themselves are small (0.09–0.89 in absolute value) and centered near zero, suggesting a genuine absence of effect rather than an underpowered trend: feature-level distillation does not yield a reliable improvement over span_tiny's main recipe, a conclusion that holds consistently across all eleven backbones tested rather than only the three originally used for this ablation.

### 5.7.2 Saliency-weighted identity-critical loss

This mechanism reweights the pixel-reconstruction loss by a saliency map derived from identity-relevant image regions, intended to concentrate SR fidelity where it matters most for recognition. Table 26 (Supplementary Material) compares the best-performing saliency weight identified during screening against the main recipe.

Again no backbone reaches significance, with small-to-moderate effect sizes (0.05–1.17 in absolute value) and a positive trend in 9/11 backbones that never clears Bonferroni correction; RegNetY-400MF comes closest ($p_{\mathrm{Bonf}}$=0.178) but is still far from significant. As with KD v2, the saliency-weighted loss does not produce a statistically detectable improvement on any of the eleven backbones tested.

### 5.7.3 Learned/differentiable block pruning

Rather than span_tiny's fixed choice to keep the first three of six SPABs, this mechanism learns which blocks to keep via a differentiable gating mechanism, with a sparsity penalty pushing the model toward a target size, in the same general family as scaling-factor-based structured pruning approaches such as Network Slimming [58], applied here at block granularity rather than per-channel. At the sparsity level that produces an exact match to span_tiny's parameter count (0.230M, 3 blocks retained), Table 27 (Supplementary Material) compares the two block-selection strategies at equal size.

Learned pruning is significantly worse than span_tiny's fixed, hand-picked block selection in 10/11 backbones, with large effect sizes throughout ($d$ −2.1 to −6.0) and every one of those ten at $p_{\mathrm{Bonf}} < 0.03$. One backbone does not fit this pattern: MobileNetV3-Large is non-significant ($d$=−0.39, $p_{\mathrm{Bonf}}$=1.0). Read as a whole, the result remains essentially unambiguous: at an identical parameter count, automatically learning which blocks to discard performs reliably worse in the large majority of backbones tested (10/11) than simply keeping the first three of six, a negative result for the differentiable-pruning mechanism specifically, not for depth compression in general (which Table 6 shows is close to accuracy-neutral when the retained blocks are chosen by hand).

Summarizing across the three: feature-level distillation and the saliency-weighted loss are statistically null in either direction across all eleven backbones, and learned pruning is significantly negative in the large majority (10/11), with the one exception (MobileNetV3-Large) non-significant rather than reversed. The paper's validated contribution remains span_tiny's original depth-count reduction (Sections 5.1–5.2).

## 5.8 Extended ablations toward mechanistic and design generality

The results above establish that depth compression is accuracy-neutral for span_tiny specifically. Three further questions were identified to test whether this finding reflects a general mechanism or an architecture-specific coincidence, and whether the specific design choices made for span_tiny are themselves well-supported. All three have now been run to completion, on EarVN1.0, to the same $n$ = 5-seed standard as the main result and extended to all eleven recognition backbones: the first (attention-parameterization) and third (block-removal position) directly, and the second (recognition-embedding-space distillation) by recognizing, on inspection, that it already has an answer from an existing result under a different name, described below.

### 5.8.1 Attention-parameterization ablation: completed, hypothesis not supported

Section 6 proposes that SPAN tolerates depth reduction because its attention mechanism carries no learned parameters, so removing blocks removes computation but not learned capacity. If this account is correct, an architecture with *learned* attention/modulation should show a measurably larger accuracy cost from an equivalent proportional depth reduction than span_tiny does. SAFMN [10] and SMFANet [11], both using learned modulation mechanisms, were each halved in block count and evaluated under the identical Track B recipe, $n = 5$ seeds, all eleven recognition backbones (SR-quality rows in Table 10). Table 28 (Supplementary Material) reports, per backbone, $\Delta_{\text{arch}}$ (that architecture's own full-depth-minus-half-depth accuracy) against $\Delta_{\text{SPAN}}$ (span_large minus span_tiny, from Table 6), and a paired test on their difference.

$\Delta_{\text{SPAN}}$ itself ($-0.0021$–$0.0114$ across backbones, Table 6) is small throughout, and neither SAFMN nor SMFANet loses significantly more than SPAN does from an equivalent proportional depth reduction in *any* of the 22 backbone×architecture combinations tested, all eleven backbones included ($p_{\text{Bonf}}$=1.0 throughout). If anything, the diff-of-diffs is negative (SPAN loses as much or more) in 12/22 combinations: SAFMN and SMFANet, despite using learned modulation mechanisms, tolerate an equivalent depth reduction just as well as span_tiny does, and this null result now holds with zero exceptions across the full eleven-backbone set rather than only the original five. The parameter-free-attention account proposed in Section 6 therefore finds no support here and is revised there accordingly: depth tolerance at this compression level and resolution appears to be a broader property of lightweight SR architectures under this training recipe, not a consequence specific to SPAN's parameter-free design.

### 5.8.2 Direct test of the low-frequency-signal hypothesis

The ablation above rejects one specific candidate mechanism (fixed, parameter-free attention) but does not itself provide positive evidence for the replacement account proposed in its place (Section 6): that the recognition-relevant signal in a low-resolution ear crop is concentrated in coarse, low-frequency structure a shallower network already captures. This was tested directly. An FFT-based frequency filter was applied to span_tiny's SR output, either lowpass (retaining only frequencies below a cutoff) or highpass (retaining only frequencies above it) at eight cutoff levels each, and the filtered images were evaluated with each backbone's already-trained recognition checkpoint (`recognition_sr_improved_<backbone>`) – no retraining, testing what a model trained on unfiltered images actually depends on at inference time, on the full 3480-image EarVN1.0 test set, $n = 5$ seeds, all eleven recognition backbones.

Table 23 reports the two most informative points from the sweep, matched for severity (90% of the spectrum removed in each direction): removing the lowest 10% of the spectrum (highpass, cutoff 0.1) collapses accuracy to 2.9–5.1% of the unfiltered baseline across all eleven backbones, near the chance level for this task, while retaining only the lowest 10% of the spectrum (lowpass, cutoff 0.1) preserves 22.3–33.6% of

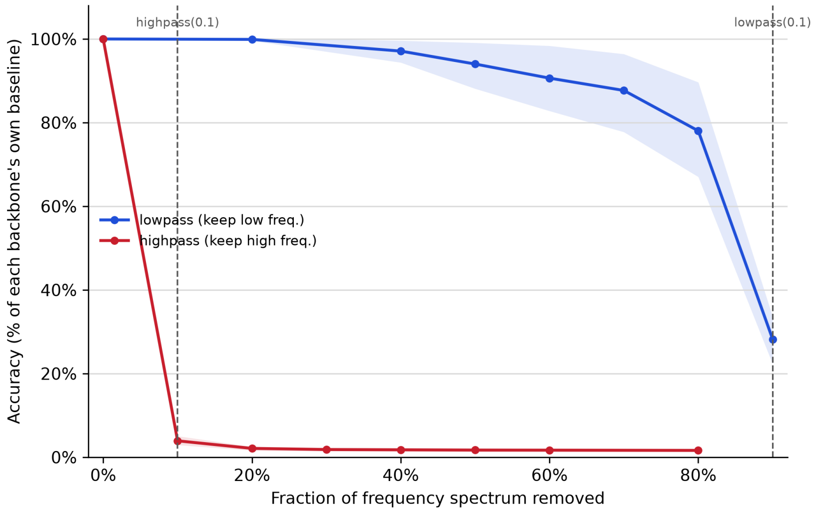



**Fig. 5** Downstream identity accuracy, normalized to each backbone's own unfiltered baseline (= 100%), as a function of the fraction of the frequency spectrum removed, EarVN1.0, $n = 5$ seeds, all eleven recognition backbones. Solid lines are the mean across all eleven backbones; shaded bands are the min–max range across backbones at each point – the asymmetry between the two curves holds with zero exceptions, on every backbone tested (per-backbone values at the two labeled cutoffs in Table 23).

**Table 23** Direct test of the low-frequency-signal hypothesis: downstream identity accuracy under FFT-based frequency filtering of span_tiny's SR output, evaluated with each backbone's already-trained recognition checkpoint (no retraining), EarVN1.0, $n = 5$ seeds, all eleven recognition backbones. "No filter" is the unfiltered baseline (verified to match exactly between the two independent no-op filter settings, lowpass cutoff= 1.0 and highpass cutoff= 0.0). "highpass (0.1)" removes the lowest 10% of the frequency spectrum; "lowpass (0.1)" keeps only the lowest 10%. Percentages are of each backbone's own baseline; the full eight-cutoff sweep underlying Figure 5 is in `results/frequency_ablation/` (code and data availability, Declarations).

| Backbone | No filter (baseline) | highpass (0.1) | lowpass (0.1) |
|---|---|---|---|
| EfficientNet-B0 | 0.5486 | 0.0230 (4.2%) | 0.1542 (28.1%) |
| GhostNet-100 | 0.5125 | 0.0174 (3.4%) | 0.1366 (26.7%) |
| LCNet-100 | 0.4843 | 0.0245 (5.1%) | 0.1392 (28.7%) |
| MobileNetV2 | 0.5018 | 0.0193 (3.8%) | 0.1118 (22.3%) |
| MobileNetV3-Large | 0.5060 | 0.0170 (3.4%) | 0.1352 (26.7%) |
| MobileNetV3-Small | 0.4413 | 0.0142 (3.2%) | 0.1185 (26.9%) |
| MobileOne-S0 | 0.4898 | 0.0220 (4.5%) | 0.1644 (33.6%) |
| RegNetY-400MF | 0.5098 | 0.0150 (2.9%) | 0.1447 (28.4%) |
| ResNet-18 | 0.4833 | 0.0229 (4.7%) | 0.1494 (30.9%) |
| ShuffleNetV2-1.0x | 0.4914 | 0.0164 (3.3%) | 0.1445 (29.4%) |
| SqueezeNet-1.1 | 0.4350 | 0.0198 (4.6%) | 0.1253 (28.8%) |

baseline accuracy – six to eight times more of each backbone's own baseline survives removing 90% of the high-frequency content than removing 90% of the low-frequency content, and this asymmetry holds with zero exceptions across all eleven backbones. The two no-op filter settings (lowpass cutoff= 1.0, highpass cutoff= 0.0), computed independently, return identical accuracy for every backbone, a built-in internal consistency check on the filtering pipeline itself. This is a direct confirmation of the low-frequency-signal account: not an inference from a rejected alternative, but a measurement of what an already-trained recognition model's own predictions actually depend on.

A complementary spectral comparison asked the converse question: does depth compression itself remove more high-frequency content than low-frequency content from the SR output? Radially-averaged power spectral density, computed on the same test set across span_tiny, span_baseline, span_large, no-SR, and HR, shows no such asymmetry – span_tiny's spectrum is nearly indistinguishable from span_baseline's and span_large's at every frequency band (summed absolute difference under 0.0003 in each half of the spectrum), smaller than the difference between any of the three SR variants and either the no-SR or HR reference. Depth compression does not measurably alter the frequency content of the SR output itself; the evidence above for the low-frequency-signal account comes from what the recognition model depends on, not from a detectable spectral signature of compression.

### 5.8.3 Block-removal position ablation: completed, no significant effect

span_tiny retains the first three of six SPABs (Figure 1), a choice made for implementation convenience rather than validated against alternatives. Two further 3-block variants, holding total block count fixed, test whether removal position matters independently of removal count: one retaining the last three blocks instead (removing blocks 1–3), and one retaining an interleaved subset (blocks 1, 3, 5).

Testing this meaningfully requires more than relabelling a freshly-initialized 3-block network: since span_tiny is trained end-to-end from a random initialization under output-level distillation only, a 3-block student trained this way has no internal notion of "which blocks of a virtual 6-block network it occupies". Three variants trained identically would be expected to differ only by seed noise, not by the position nominally assigned to them. To give position a training signal, each variant instead uses a position-targeted feature-hint loss: the student's final-block output is matched, via a learned $1 \times 1$ adapter and the same instance-normalized $L_1$ formulation already used for the feature-KD term (Section 5.7), against the teacher's output at a specific block index representing the depth that variant is meant to occupy (block 3 for the current first-three-blocks recipe, block 6 for a last-three-blocks recipe, block 5 for an interleaved 1-3-5 recipe). All three variants use this same mechanism, differing only in which teacher block is targeted, so the comparison isolates position specifically rather than confounding it with the presence or absence of positional supervision. The teacher for this ablation is span_large rather than span_baseline/span_official: both are 6-block, but only span_large shares the same re-implementation class as span_tiny (Section 3.2), giving well-defined intermediate block outputs to target;

span_official's internal structure is external, third-party code without a guaranteed matching interface.

Table 24 reports downstream identity accuracy for the three variants across all eleven recognition backbones at $n = 5$ seeds.

**Table 24** Block-removal position ablation, downstream accuracy, EarVN1.0, $n = 5$ seeds, all eleven recognition backbones.

| Backbone | keep-first (current) | keep-last | interleaved |
|---|---|---|---|
| EfficientNet-B0 | 0.5389 | 0.5417 | 0.5403 |
| GhostNet-100 | 0.5027 | 0.5084 | 0.5034 |
| LCNet-100 | 0.4721 | 0.4670 | 0.4681 |
| MobileNetV2 | 0.4837 | 0.4890 | 0.4820 |
| MobileNetV3-Large | 0.5034 | 0.4970 | 0.5038 |
| MobileNetV3-Small | 0.4246 | 0.4240 | 0.4330 |
| MobileOne-S0 | 0.4821 | 0.4728 | 0.4756 |
| RegNetY-400MF | 0.4946 | 0.4983 | 0.4974 |
| ResNet-18 | 0.4763 | 0.4791 | 0.4740 |
| ShuffleNetV2-1.0x | 0.4733 | 0.4718 | 0.4775 |
| SqueezeNet-1.1 | 0.4167 | 0.4222 | 0.4240 |

None of the three pairwise comparisons reaches significance in any backbone ($p_{\text{Bonf}}$ 0.15–1.0 across all 33 backbone$\times$pair combinations), with effect sizes small throughout. Extending from five to eleven backbones changes the descriptive picture, though not the statistical one: keep-last ranked highest in $4/5$ of the original backbones, a pattern that does not hold up as a majority once the six new backbones are included – keep-last is highest in $5/11$, keep-first in $2/11$ (LCNet-100, MobileOne-S0), and the interleaved variant in $4/11$ (MobileNetV3-Large, MobileNetV3-Small, ShuffleNetV2-1.0x, SqueezeNet-1.1). Since none of these differences was ever statistically significant, this is best read as confirming there was no real directional trend to begin with, only sampling noise in which variant happened to rank first among a small set of backbones; the null result on removal position itself is unaffected and, if anything, more convincingly supported by the more even split across eleven backbones than by the original apparent (but never significant) lean toward keep-last. Block-removal position does not detectably affect downstream accuracy at this compression level, on any of the eleven backbones tested.

An initial $n = 3$-seed screening pass, run before extending to the full five seeds, had flagged one nominally significant difference (EfficientNet-B0, keep-first vs. interleaved: $d = 8.80$, $p_{\text{Bonf}} = 0.013$). It did not replicate once seeds 44 and 999 were added ($d = -0.13$, $p_{\text{Bonf}} = 1.0$): the added seeds shifted keep-first's mean downward and widened its variance enough to erase the effect entirely. This is the extension protocol working as intended: the same $n = 3 \to n = 5$ seed-extension rule applied throughout this study (Section 5.7) exists specifically to catch small-sample false positives of this kind before they are reported as findings.

### 5.8.4 Recognition-embedding-space distillation

Ataer-Cansizoglu et al. [12] and the iris-recognition literature reviewed in Section 2 (Nguyen et al. [30], Ribeiro et al. [31]) suggest that SR objectives aligned with a downstream recognition embedding, rather than with pixel or teacher-feature reconstruction, are more likely to transfer to recognition accuracy. This account is directly testable with a loss term matching embeddings extracted from a frozen, pretrained recognition backbone, computed between span_tiny's SR output and the true HR image:

$$\mathcal{L}_{\text{embed}} = 1 - \cos\big(f(\text{SR}(x)),\, f(\text{HR})\big), \tag{1}$$

where $f(\cdot)$ denotes the frozen backbone's penultimate-layer embedding. This is, on inspection, mathematically identical to the identity-loss term already evaluated in Section 5.3 and Section 3.3: both are a cosine loss between a frozen recognition backbone's embeddings of the SR output and the true HR image, using the same backbone (MobileNetV2) as the embedding source. The two configurations this account calls for (pixel plus $\mathcal{L}_{\text{embed}}$ alone, and pixel plus distillation plus $\mathcal{L}_{\text{embed}}$) are exactly the pixel+identity and pixel+distill+identity ("full") rows already reported in Table 9, not a new experiment. Both fall in the lower-performing cluster there on MobileNetV2, corroborated by the standalone $\lambda_{\text{identity}}$ sweep in Section 3.3 on the same backbone. The recognition-aware-objective account therefore does not hold in this setting: unlike the face- and iris-recognition results that motivate it, aligning the SR loss with a frozen recognition backbone's embedding space does not improve, and in fact measurably harms, downstream identity accuracy for ear SR at this compression level.

## 6 Discussion

The design goal of this work was not maximal downstream accuracy but a specific, pre-defined criterion: preserve the ordering `lr` < `span_tiny` < `span_baseline` while substantially reducing parameter count and latency, and in particular keep the compression-specific cost (`span_tiny` vs. `span_baseline`) small and, ideally, confirmed negligible. On EarVN1.0 this cost is non-significant in 10/11 backbones (Table 4), and TOST equivalence testing confirms 1 of those 10 as negligible at $n = 5$ (ShuffleNetV2-1.0x); the depth-isolated comparison (Table 6) gives the cleaner answer, non-significant in 10/11 backbones and TOST-confirmed equivalent in 3 of those 10, with the remaining 7 inconclusive rather than settled. On AWEx it is non-significant in 11/11 backbones under all three protocols tested (from-scratch, full-fine-tune transfer, and frozen-backbone transfer; Section 5.6), the single most consistent result in the paper by the non-significance criterion alone; TOST tells a more modest story there, however, confirming genuine equivalence in only 2 of the 33 backbone×protocol combinations (both on the from-scratch track), with the remainder non-significant but not confirmed negligible. This forms the paper's central empirical claim: no detected cost almost everywhere, confirmed negligible cost in a subset, and an open question in the rest at this sample size.

Section 3.2 proposed that SPAN tolerates depth reduction because its attention mechanism carries no learned parameters: removing SPABs removes attention *computation* but not learned attention *capacity*, since there is none to lose. This account makes a testable directional prediction (an architecture with *learned* attention/modulation should lose more from an equivalent depth cut), and Section 5.8 tested it directly against SAFMN and SMFANet. The prediction was not confirmed: neither architecture loses significantly more than SPAN does from a proportional depth halving in any of 22 backbone×architecture combinations (all eleven recognition backbones, both SAFMN and SMFANet), and the diff-of-diffs is negative (SPAN losing as much or more) in 12/22. The parameter-free-attention account is therefore not supported by this test and is retracted as the explanation. A more likely account, consistent with both this ablation and the loss-ablation result below, is that at this extreme compression level and input resolution (20×20 px), the recognition-relevant signal in an ear crop is concentrated in relatively coarse, low-frequency structure that a shallower network already captures adequately, regardless of its attention mechanism's parameterization, making depth reduction broadly tolerable across this class of lightweight SR architectures rather than a property specific to SPAN. Unlike the parameter-free-attention account it replaces, this one was tested directly rather than left as an untested hypothesis (Section 5.8.2): frequency-filtering span_tiny's SR output and re-evaluating each backbone's already-trained recognition checkpoint shows accuracy collapsing to 2.9–5.1% of baseline when the lowest 10% of the spectrum is removed, but retaining 22.3–33.6% of baseline when only the lowest 10% is kept, with zero exceptions across all eleven backbones – direct evidence of what the recognition model depends on, rather than an inference from a rejected alternative. A complementary spectral comparison, however, finds no corresponding difference in frequency content between the SR outputs of compressed and uncompressed models themselves, so this account is supported as a description of the recognition model's own input dependence, not as a demonstrated property of what depth compression removes from the image.

Section 5.3 finds two distinct results that should not be conflated: on MobileNetV2, the identity-loss term *significantly reduces* downstream accuracy (3/6 pairwise comparisons significant, 1/6 at trend level, corroborated independently by the $\lambda_{\text{identity}}$ sweep in Section 3.3 on the same backbone); extending this ablation to ten further backbones (eleven in total) shows the effect is backbone-dependent rather than universal, echoed more weakly (one at trend level only) on two of the ten (ShuffleNetV2-1.0x, LCNet-100), and undetectable on the remaining eight, so it should be read as a real but backbone-dependent finding rather than either a MobileNetV2-specific artifact or a general property of the mechanism. The distillation term's own marginal contribution, by contrast, remains statistically indistinguishable from pixel-only training across all eleven backbones tested, TOST-confirmed equivalent on four of them (EfficientNet-B0, GhostNet-100, RegNetY-400MF, SqueezeNet-1.1). For the latter, null result, three explanations seem plausible: the pixel-reconstruction signal alone may already be close to sufficient at this parameter scale; seed-to-seed variance may exceed the true effect size; or, as Zhang et al. [35] argue for SR distillation generally, a teacher's own output is only an approximation of the true HR image, which caps how much signal the distillation term can supply. The data collected here cannot distinguish between these. Neither result undermines the paper's central claim, which concerns architectural depth, not loss-function choice; span_tiny's training recipe (Track B, pixel+distillation, $\lambda_{\text{identity}}$=0) remains justified independently by its role in stabilizing convergence for other architectures in the contextual comparison (Section 5.4) and by the identity-loss ablation's own negative finding for that specific term.

SMFANet failed to converge well under pixel-loss-only supervision (20.906 dB, Table 10) but recovered strongly under distillation supervision (+5.4 dB). The marginal contribution of distillation to span_tiny's own downstream accuracy was not statistically detectable, but this case shows that distillation supervision is not uniformly redundant across architectures at this extreme input resolution. Some architectures appear to depend on it to converge at all.

Beyond the SPAN family, span_tiny is second-fastest and third-smallest among six compared architectures, and, once all four contextual architectures are measured at the same $n = 5$ SR-training-seed standard as span_tiny (Table 10), is statistically indistinguishable from RLFN-adapted on PSNR/SSIM, both behind only SAFMN and both clearly ahead of ECBSR/SMFANet – not top-ranked, but no longer accurately described as low-ranked either. On downstream recognition accuracy, the metric this paper is organized around, span_tiny is never significantly beaten by any of the other four architectures across any of the eleven recognition backbones, is statistically tied with the two most accurate ones (RLFN-adapted, SAFMN), and significantly *outperforms* the two least accurate ones in most backbones (ECBSR: 8/11 significant, 1/11 trend, 2/11 tied; SMFANet: 4/11 significant, 3/11 trend, 4/11 tied) (Table 11). This PSNR/SSIM grouping and this accuracy grouping now largely agree (SAFMN/RLFN-adapted/span_tiny cluster together on both, ECBSR/SMFANet trail on both), so the sharper point is not that raw image quality misidentifies span_tiny specifically, but that it is worth stating plainly what span_tiny's position does and does not mean: it is not evidence that span_tiny is the best lightweight SR architecture in absolute terms (SAFMN and RLFN-adapted remain fully competitive, at higher latency for SAFMN and a comparable one for RLFN-adapted), but it is evidence that a compressed architecture can match top-performing, larger alternatives on the metric that actually matters for this task without matching or exceeding them on pixel fidelity. The re-implementation's timing broadly matches the original SPAN paper's comparison against SAFMN on standard SR benchmarks ($\approx$13 ms/28.66 dB vs. $\approx$72 ms/$\approx$28.60 dB on Set14 $\times 4$ in [5]), an independent data point supporting the general efficiency trade-off observed here, alongside the new evidence that this trade-off does not come at a downstream-accuracy cost relative to the weaker-PSNR alternatives.

The weaker and more protocol-dependent statistical significance of `SR` > `lr` on AWEx (Section 5.6) compared to EarVN1.0 (Table 4, significant in all eleven backbones for both span_tiny and span_baseline) follows from AWEx's smaller per-subject training sample ($\approx$6.7 images/subject, versus substantially more on EarVN1.0), not from a weaker or contradictory effect. It is not evidence of no effect. This applies specifically to the *secondary* question (does SR help over no-SR); the paper's *central* claim, that compression itself is safe, is the more consistent finding across both datasets and is confirmed under all three protocols tested on AWEx with 0/11 backbones showing a significant cost in any of them; if anything, this claim is *more* thoroughly replicated on AWEx (three independent protocols) than the secondary question is.

Two anomalies flagged at an earlier stage of this study are resolved under the corrected experimental setup and are not carried forward: an off-trend ResNet-18 result on the AWEx from-scratch track (Section 5.6), and span_large's anomalously low PSNR on AWEx (previously ≈19 dB; now 24.4 dB, in line with the other three core architectures, Table 20). One new finding is reported as an investigated boundary condition rather than an unexplained anomaly: a backbone-dependent reversal of the ordering on the EarVN1.0 real-low-resolution holdout (Section 5.5), which reaches statistical significance on only one of the eleven backbones tested (ResNet-18) while two more show it non-significantly and the rest do not, and is traced to every holdout image's native resolution falling below the SR models' trained input size – an artifact some recognition backbones are evidently more robust to than others. One genuine open item remains: ResNet-18's inconsistent secondary-question evidence across all three AWEx protocols (Section 5.6.2), for which no confirmed explanation exists and which is carried forward as a limitation rather than a subject for speculation.

# 7 Limitations

Several limitations concern AWEx specifically, and none of them affects the paper's central compression-safety claim, which is confirmed there with zero of eleven backbones showing a significant cost under all three protocols tested. AWEx's smaller per-subject sample size (≈6.7 training images/subject) does limit statistical power for the secondary question of whether SR helps over no-SR at all (Section 5.6): effects are directionally positive in most backbone×protocol combinations but do not always clear the significance threshold, and which backbones reach significance varies by protocol; this should be read as underpowered rather than null (Section 6). ResNet-18 is the clearest case of this: across all three AWEx protocols it shows the weakest and least consistent evidence that SR helps at all, including one instance (frozen-backbone transfer) where span_tiny falls nominally, non-significantly, below no-SR. No confirmed explanation exists for why this backbone behaves differently from the other ten on this dataset, but it is notable that the same backbone also shows the only statistically solid instance of an analogous reversal on EarVN1.0's real-low-resolution holdout (Section 5.5), a general-fragility-to-distribution-shift account offered there as a candidate, not confirmed, explanation for both observations together. The AWEx depth ablation (span_tiny vs. span_large) also runs at a reduced sample size for the five original backbones, $n = 3$ seeds rather than the $n = 5$ used elsewhere (the six backbones added later use the standard $n = 5$ on this ablation), so its non-significant result across all eleven backbones carries correspondingly less power for the original five, even though it agrees with the EarVN1.0 finding. Separately, the six-architecture context comparison, the attention-parameterization ablation, and the three additional-mechanism ablations were all validated on EarVN1.0 only in this phase of the study; whether span_tiny's competitive standing against other architectures, the null result for the parameter-free-attention account, and the negative or null findings for KD v2, the saliency-weighted loss, and learned pruning replicate on AWEx is untested and left as a direction for future work. Finally, AWEx is evaluated with a custom closed-set, per-subject train/validation/test split rather than its official open-set split, because this study addresses closed-set identity classification rather than the open-set verification problem that official split was designed for; this choice is considered appropriate for the task at hand and is stated explicitly to avoid any impression of an undisclosed deviation from a standard protocol.

The loss-component ablation (Section 5.3) has been extended to all eleven backbones, and this full coverage refines the finding's interpretation: the identity-loss term's significant cost is clearest on MobileNetV2, echoed more weakly on two further backbones (ShuffleNetV2-1.0x at trend level, LCNet-100 at significance on one comparison), and undetectable on the remaining eight backbones tested (EfficientNet-B0, GhostNet-100, MobileNetV3-Large, MobileNetV3-Small, MobileOne-S0, RegNetY-400MF, ResNet-18, SqueezeNet-1.1), so it should be read as a real but backbone-dependent finding rather than a general property of the mechanism. Whether this pattern holds on AWEx remains untested, and the independent $\lambda_{\text{identity}}$ sweep (Section 3.3) still covers MobileNetV2 only. The unmodified `rlfn` baseline (distinct from `rlfn_adapted`, the primary RLFN reference used throughout) has $n = 3$ seeds at the time of writing, short of the $n = 5$ used elsewhere, and remains validated on the original five backbones only.

Two measurement-level limitations apply throughout. Latency figures (Table 10) come from the development machine rather than a representative resource-constrained device such as a Raspberry Pi or a phone SoC, so they should be read as relative comparisons under controlled conditions rather than deployment-ready benchmarks. PSNR/SSIM/LPIPS for the three core SPAN-family variants (span_baseline, span_tiny, span_large) are reported as mean±std across the same $n = 5$ SR-training seeds used for the SR-seed-variance analysis below, on both EarVN1.0 (Table 10) and AWEx (Table 20), reusing checkpoints already trained for that purpose rather than requiring new experiments; this closes the $n = 1$-vs-$n = 5$ asymmetry with the downstream-accuracy tables for the paper's central architecture comparison specifically, on both datasets. On EarVN1.0, this has since been extended further: the EDSR teacher and the four Track-B contextual architectures (RLFN-adapted, ECBSR, SAFMN, SMFANet) now also report mean±std at the same $n = 5$, reusing the same seed-variance infrastructure rather than new experiments (Table 10). What remains open: on AWEx, the EDSR teacher is still $n = 1$ (Table 20; the contextual-comparison architectures were never run on AWEx at all, at any $n$); and on EarVN1.0, the Track-A versions of all five contextual architectures (RLFN, RLFN-adapted, ECBSR, SAFMN, SMFANet) and the two half-depth rows (Table 10) were never retrained at multiple SR seeds and remain $n = 1$, with no confidence interval – these rows are secondary reference points (matching the original papers' own recipes, or supporting a separate ablation) rather than part of this paper's central Track-B comparison. The related question of whether the *downstream* multi-seed results depend on which SR-training seed happened to be used was checked directly, across all eleven recognition backbones and all three architectures rather than a single screening backbone (Section 3.6): SR-training-seed variance exceeds the already-reported downstream-recognition-seed variance in 26 of 33 backbone/architecture combinations, confirmed by a crossed mixed-effects model, with a complete 25-cell SR-seed×downstream-seed grid per backbone×domain, on Table 4's central comparison across all eleven backbones (20.2% of variance attributed to SR-training seed, versus an estimated 0.0% for downstream-recognition seed, the latter pushed fully to the boundary of the parameter space under REML, Section 3.6). This is not specific to span_baseline (span_tiny itself shows a comparable or larger ratio on three of eleven backbones), so it is reported as a general property of this pipeline's SR-training stochasticity rather than a caveat attached only to the comparison baseline. It does not overturn any conclusion in this paper, since the main comparisons' point estimates are unaffected, but it does mean the reported downstream-seed standard deviations understate the total uncertainty attributable to SR-training choices, and a reader treating the reported confidence intervals as capturing all sources of variance should read this section's ratios as a correction to that assumption.

Multiple-comparison correction throughout this paper uses Bonferroni [49] rather than a less conservative alternative such as Holm-Bonferroni [59], which controls the same family-wise error rate with uniformly higher power. Bonferroni was retained deliberately: it is simpler to verify by hand from a single multiplier ($p_{\mathrm{Bonf}} = \min(1, p_{\mathrm{raw}} \times n_{\mathrm{comparisons}})$) than a sequential procedure, and, since every reported $p_{\mathrm{Bonf}}$ in this paper is already conservative, switching corrections could in principle move some non-significant comparisons into significance but never the reverse. This paper's non-significant findings are therefore, if anything, understated rather than overstated by this choice, and its significant findings would remain significant under the less conservative alternative.

Finally, the paper's main claim is validated specifically for the synthetic bicubic-downsampling regime defined in Section 3, and this scope is not assumed to extend further: an auxiliary check on genuinely low-resolution real-world images (Section 5.5) shows the opposite ordering on some recognition backbones, traced to a resolution mismatch between these images (all natively below the SR models' trained input size) and the training-time degradation. This is a boundary condition on where the claim has been shown to hold, not a refutation of it, but it means this paper does not provide evidence that span_tiny or span_baseline improves recognition on arbitrary real-world low-resolution ear images outside the synthetic-degradation regime studied here. An initial mobilenet_v2-only screen was extended to all eleven recognition backbones at $n = 5$ seeds each on this same 173-image holdout (Section 5.5), which showed the reversal is not a general property of this regime: it reaches statistical significance on only one backbone (ResNet-18), is a non-significant nominal reversal on two more, and the remaining backbones show either no reversal or SR nominally *helping* on this same holdout, at trend level for two of them (MobileNetV3-Small, LCNet-100) – evidence that this boundary condition is backbone-dependent rather than a property of the real-low-resolution images or the SR reconstructions themselves. Gender accuracy on this holdout remains a mobilenet_v2-only measurement not yet extended to the other ten backbones; since identity and gender labels are derived from the same per-subject image ranges on the same 173 images, this is not independent evidence from the identity-accuracy result above. The degradation-augmented mitigation (Section 5.5.1) was subsequently extended to four further backbones (ResNet-18, EfficientNet-B0, GhostNet-100, ShuffleNetV2-1.0x), including the one backbone where the reversal is actually statistically solid: on ResNet-18, the mitigation does not help – span_baseline remains significantly worse than no-SR on both identity and gender accuracy under the degradation-augmented recipe, just as under the original recipe – while two of the other three backbones (GhostNet-100, ShuffleNetV2-1.0x) show no reversal at all under this recipe, with SR helping significantly instead, and the fourth (EfficientNet-B0) shows no significant effect in either direction. The mitigation therefore remains partial and backbone-dependent rather than a general fix, but is no longer an untested question for the specific backbone where the underlying problem is most solidly established; the remaining six backbones are untested.

# 8 Conclusion

This paper asked a narrower question than most efficient-SR work: not whether a compressed model is the most accurate option available, but whether compressing SPAN's depth by half still leaves a model worth using ahead of no SR at all, without paying a real accuracy price for the compression itself. Across eleven recognition backbones and two independent datasets, span_tiny answers both parts of that question affirmatively. `span_tiny` beats `no-SR` with statistical significance in all eleven backbones on EarVN1.0; the specific cost of compression (`span_tiny` vs. `span_baseline`) is non-significant in 10/11 backbones there and in 11/11 backbones under all three evaluation protocols tested on AWEx (from-scratch, full-fine-tune transfer, frozen-backbone transfer), the most consistently replicated finding in the paper; TOST confirms genuine equivalence for 1 of the 10 non-significant EarVN1.0 backbones individually at $n = 5$, and, averaged across all eleven backbones with both the SR-training-seed and downstream-recognition-seed variance sources modeled jointly (Section 3.6) rather than one held fixed, the compression cost is confirmed negligible at the pooled level too (TOST, $p = 0.008$). The depth ablation isolating block count alone gives the cleaner answer: span_tiny and its uncompressed-depth counterpart, span_large, are statistically indistinguishable in 10/11 backbones on EarVN1.0 and 11/11 on AWEx, and TOST confirms genuine equivalence in 3 of the 10 non-significant EarVN1.0 backbones, leaving the other 7 open rather than settled. A full depth sweep across every intermediate block count, extended from an initial screening backbone to five spanning distinct architecture families, locates the compression boundary itself rather than testing a single cut: the most aggressive 1-block cut produces a significant cost in three of the five, but the confirmed floor of safe compression is backbone-dependent rather than a single number – span_tiny's own 3-block choice is TOST-confirmed equivalent to the 6-block reference for two of the five (MobileNetV2, GhostNet-100), no floor is confirmed anywhere across the entire 2–5 block range for two more (ResNet-18, ShuffleNetV2-1.0x), and the fifth (EfficientNet-B0) shows a significant cost specifically at the 3-block point despite both fewer and more blocks being safe, a non-monotonic pattern rather than a clean depth-vs-accuracy trend. This directly answers how far this architecture can be compressed rather than only whether one particular cut is safe, at the cost of showing that answer is itself backbone-dependent. Against four other lightweight SR architectures, span_tiny is never significantly beaten on downstream accuracy and significantly outperforms the two weakest of them in most backbones, ties for downstream accuracy with the two strongest despite matching, not exceeding, their PSNR/SSIM at a fraction of their parameter count – a concrete demonstration that pixel-fidelity ranking is not what determines recognition-relevant SR quality at this compression level.

Three further mechanisms proposed to improve on span_tiny (a richer distillation loss, a saliency-weighted loss, and learned block pruning) show no improvement: two are statistically null in either direction, and learned pruning is significantly worse than span_tiny's fixed block selection in 10 of 11 backbones tested (MobileNetV3-Large non-significant). An ablation designed to test this paper's own proposed explanation for why depth reduction is tolerable (SPAN's parameter-free attention) returned a result against that explanation. A direct frequency-domain test of the replacement account – that recognition depends on coarse, low-frequency structure – found strong, consistent support instead: removing the lowest 10% of the input spectrum collapsed downstream accuracy to 2.9–5.1% of baseline in all eleven backbones, while retaining only that same lowest 10% preserved 22.3–33.6% of baseline, with zero exceptions across backbones; a complementary spectral comparison, however, found no corresponding difference in frequency content between the SR outputs of compressed and uncompressed models themselves, so this evidence concerns what the recognition model depends on rather than what compression removes. An auxiliary check on genuinely real, low-resolution images reversed the main claim's ordering on some recognition backbones but not others (statistically solid for only one of eleven tested, ResNet-18), traced to a specific, verifiable cause: a resolution mismatch between these images and the synthetic degradation used in training, one some backbones are more robust to than others – a boundary condition on the paper's scope. A direct mitigation for this boundary condition, degradation-augmented retraining, was tested on an initial screening backbone and, subsequently, on four further backbones: it helps partially and asymmetrically on the screening backbone (narrowing or reversing span_tiny's gap to no-SR without disturbing the compression-safety claim, though not to the point of statistical significance, and without the same benefit for span_baseline), helps two of the four further backbones significantly with no reversal at all under the augmented recipe, shows no effect on a third, and does not rescue the fourth (ResNet-18, the one backbone where the original reversal is statistically solid) – span_baseline remains significantly worse than no-SR there even under the augmented recipe. One anomaly remains open: ResNet-18's inconsistent secondary-question evidence on AWEx, across all three protocols, lacks a confirmed explanation. None of this changes the central result.

Extending this work would involve latency measured on an actual edge device rather than development hardware, confidence intervals for the SR-quality metrics still measured at a single run – the Track-A and half-depth rows in Table 10 (Section 7), now that EDSR and the four Track-B contextual architectures have joined span_baseline/span_tiny/span_large at $n = 5$ – extending the six-architecture context comparison and the three additional-mechanism ablations to AWEx to test whether they replicate cross-dataset, and, if a suitable third dataset becomes available, an additional independent check on generalization beyond EarVN1.0 and AWEx. Two further questions raised in Section 5.8 about span_tiny's specific design choices have since been resolved rather than left open: a block-removal position ablation, testing whether the specific choice of retained blocks was well-supported, found no significant effect of removal position at $n = 5$ seeds across all eleven backbones; and a candidate recognition-embedding-space distillation loss, motivated by the recognition-aware SR literature reviewed in Section 2, turned out on inspection to be mathematically identical to the identity-loss term already evaluated in Section 5.3, an account that does not hold in this setting. A further question about the multi-seed protocol itself (whether fixing the SR-training seed and repeating only the downstream-recognition seed could understate the paper's true variance) was also checked directly rather than assumed away, across all eleven backbones and all three architectures, with a complete crossed SR-seed$\times$downstream-seed grid for every backbone, and confirmed with a crossed mixed-effects model on the paper's central comparison rather than a descriptive standard-deviation ratio alone (Section 3.6): SR-training-seed variance exceeds the already-reported downstream-recognition-seed variance in most backbone/architecture combinations, including for span_tiny, so this is reported as a general property of the pipeline rather than a caveat specific to the comparison baseline.

The four ablations below test mechanisms considered as candidate improvements to span_tiny's design but not adopted in the final recipe; each is discussed and interpreted in the main text (Sections 5.7–5.8), with only the full per-backbone table moved here to keep the main text within length.

# Appendix A Feature-level distillation ("KD v2")

No backbone shows a statistically significant difference from span_tiny's main recipe in either direction ($p_{\mathrm{Bonf}}$ 0.35–1.0), and the sign of the effect is inconsistent across backbones (positive in 5/11, negative in 6/11); Table 25 reports the full per-backbone results underlying this null finding (Section 5.7).

**Table 25** KD v2 ablation vs. span_tiny's main recipe, downstream accuracy, EarVN1.0, $n = 5$ seeds, all eleven recognition backbones.

| Backbone | span_tiny | span_tiny+KDv2 | Cohen's $d$ | $p_{\mathrm{Bonf}}$ |
|---|---|---|---|---|
| EfficientNet-B0 | 0.5485 | 0.5512 | 0.26 | 1.0 |
| GhostNet-100 | 0.5115 | 0.5142 | 0.30 | 1.0 |
| LCNet-100 | 0.4852 | 0.4869 | 0.09 | 1.0 |
| MobileNetV2 | 0.5026 | 0.4906 | −0.89 | 0.3542 |
| MobileNetV3-Large | 0.5072 | 0.4999 | −0.65 | 0.657 |
| MobileNetV3-Small | 0.4423 | 0.4352 | −0.42 | 1.0 |
| MobileOne-S0 | 0.4898 | 0.4873 | −0.15 | 1.0 |
| RegNetY-400MF | 0.5107 | 0.5017 | −0.75 | 0.503 |
| ResNet-18 | 0.4825 | 0.4848 | 0.69 | 0.5992 |
| ShuffleNetV2-1.0x | 0.4928 | 0.4889 | −0.42 | 1.0 |
| SqueezeNet-1.1 | 0.4352 | 0.4407 | 0.19 | 1.0 |

# Appendix B Saliency-weighted identity-critical loss

No backbone reaches significance against span_tiny's main recipe, with small-to-moderate effect sizes (0.05–1.17 in absolute value) and a positive trend in 9/11 backbones that never clears Bonferroni correction; RegNetY-400MF comes closest ($p_{\mathrm{Bonf}}$=0.178) but is still far from significant. Table 26 reports the full per-backbone results (Section 5.7).

**Table 26** Saliency-weighted loss ablation vs. span_tiny's main recipe, downstream accuracy, EarVN1.0, $n = 5$ seeds, all eleven recognition backbones.

| Backbone | span_tiny | span_tiny+saliency | Cohen's $d$ | $p_{\mathrm{Bonf}}$ |
|---|---|---|---|---|
| EfficientNet-B0 | 0.5485 | 0.5559 | 0.90 | 0.3462 |
| GhostNet-100 | 0.5115 | 0.5150 | 0.42 | 1.0 |
| LCNet-100 | 0.4852 | 0.4866 | 0.08 | 1.0 |
| MobileNetV2 | 0.5026 | 0.5053 | 0.29 | 1.0 |
| MobileNetV3-Large | 0.5072 | 0.5373 | 0.75 | 0.502 |
| MobileNetV3-Small | 0.4423 | 0.4441 | 0.11 | 1.0 |
| MobileOne-S0 | 0.4898 | 0.4821 | −0.73 | 0.536 |
| RegNetY-400MF | 0.5107 | 0.5198 | 1.17 | 0.178 |
| ResNet-18 | 0.4825 | 0.4800 | −0.29 | 1.0 |
| ShuffleNetV2-1.0x | 0.4928 | 0.4934 | 0.05 | 1.0 |
| SqueezeNet-1.1 | 0.4352 | 0.4364 | 0.10 | 1.0 |

# Appendix C Learned/differentiable block pruning

Learned pruning is significantly worse than span_tiny's fixed, hand-picked block selection in 10/11 backbones at an identical parameter count, with large effect sizes throughout ($d$ −2.1 to −6.0); the one exception, MobileNetV3-Large, is non-significant ($d$=−0.39, $p_{\mathrm{Bonf}}$=1.0) rather than reversed. Table 27 reports the full per-backbone results (Section 5.7).

**Table 27** Learned block pruning vs. span_tiny's fixed block selection, downstream accuracy at an identical parameter count (0.230M), EarVN1.0, $n = 5$ seeds, all eleven recognition backbones.

| Backbone | span_tiny (fixed) | Learned pruning | Cohen's $d$ | $p_{\text{Bonf}}$ |
|---|---|---|---|---|
| EfficientNet-B0 | 0.5485 | 0.5222 | −3.71 | **0.0035** |
| GhostNet-100 | 0.5115 | 0.4806 | −3.54 | **0.0041** |
| LCNet-100 | 0.4852 | 0.4546 | −3.79 | **0.0032** |
| MobileNetV2 | 0.5026 | 0.4685 | −3.00 | **0.0078** |
| MobileNetV3-Large | 0.5072 | 0.4999 | −0.39 | 1.0 |
| MobileNetV3-Small | 0.4423 | 0.4095 | −3.34 | **0.0052** |
| MobileOne-S0 | 0.4898 | 0.4579 | −2.11 | **0.0274** |
| RegNetY-400MF | 0.5107 | 0.4726 | −3.26 | **0.0056** |
| ResNet-18 | 0.4825 | 0.4575 | −5.98 | **0.0005** |
| ShuffleNetV2-1.0x | 0.4928 | 0.4602 | −2.78 | **0.0102** |
| SqueezeNet-1.1 | 0.4352 | 0.4032 | −3.37 | **0.0050** |

# Appendix D Attention-parameterization ablation

Neither SAFMN nor SMFANet loses significantly more than SPAN does from an equivalent proportional depth reduction in any of the 22 backbone×architecture combinations tested (all eleven recognition backbones, both architectures), and the diff-of-diffs is negative (SPAN losing as much or more) in 12/22 combinations, so the parameter-free-attention account is not supported (Section 5.8). Table 28 reports the full per-backbone results.

**Table 28** Attention-parameterization ablation, EarVN1.0, $n = 5$ seeds, all eleven recognition backbones. $\Delta_{\text{arch}} - \Delta_{\text{SPAN}}$ is the diff-of-diffs (positive = the learned-attention architecture loses *more* from depth-halving than SPAN, supporting the parameter-free account); $p_{\text{Bonf}}$ corrected across 11 backbones per architecture.

| Backbone | $\Delta_{\text{SAFMN}}$ | $\Delta_{\text{SMFANet}}$ | diff-of-diffs ($d$, $p_{\text{Bonf}}$) SAFMN | diff-of-diffs ($d$, $p_{\text{Bonf}}$) SMFANet |
|---|---|---|---|---|
| EfficientNet-B0 | 0.0066 | 0.0023 | −0.0017, $d$=−0.14, $p$=1.0 | −0.0060, $d$=−0.79, $p$=1.0 |
| GhostNet-100 | −0.0021 | 0.0012 | −0.0046, $d$=−0.54, $p$=1.0 | −0.0013, $d$=−0.18, $p$=1.0 |
| LCNet-100 | 0.0064 | −0.0030 | −0.0050, $d$=−0.23, $p$=1.0 | −0.0144, $d$=−0.67, $p$=1.0 |
| MobileNetV2 | 0.0019 | 0.0002 | 0.0000, $d$=0.00, $p$=1.0 | −0.0016, $d$=−0.14, $p$=1.0 |
| MobileNetV3-Large | 0.0188 | 0.0017 | 0.0089, $d$=0.22, $p$=1.0 | −0.0083, $d$=−0.24, $p$=1.0 |
| MobileNetV3-Small | −0.0009 | 0.0074 | −0.0059, $d$=−0.33, $p$=1.0 | 0.0025, $d$=0.30, $p$=1.0 |
| MobileOne-S0 | −0.0088 | −0.0000 | −0.0027, $d$=−0.20, $p$=1.0 | 0.0060, $d$=0.43, $p$=1.0 |
| RegNetY-400MF | 0.0003 | 0.0032 | −0.0037, $d$=−0.33, $p$=1.0 | −0.0008, $d$=−0.07, $p$=1.0 |
| ResNet-18 | 0.0038 | −0.0008 | 0.0000, $d$=0.00, $p$=1.0 | −0.0046, $d$=−0.33, $p$=1.0 |
| ShuffleNetV2-1.0x | 0.0083 | −0.0018 | 0.0088, $d$=0.61, $p$=1.0 | −0.0013, $d$=−0.08, $p$=1.0 |
| SqueezeNet-1.1 | 0.0122 | 0.0013 | 0.0144, $d$=0.80, $p$=1.0 | 0.0035, $d$=0.29, $p$=1.0 |

# Appendix E Depth sweep: full pairwise comparisons

Table 8 in the main text (Section 5.2.1) reports a verdict summary only. Table 29 below reports the full effect sizes, Bonferroni-corrected $p$-values, and TOST $p$-values underlying that summary.

**Table 29** Each block count vs. the 6-block (uncompressed) reference, EarVN1.0, $n = 5$ seeds, five backbones – full statistics for Table 8. $p_{\mathrm{Bonf}}$ corrected across the 5-comparison family reported by each backbone's own aggregation (Section 3.5).

| Backbone | 1 vs. 6 | 2 vs. 6 | 3 vs. 6 | 4 vs. 6 | 5 vs. 6 |
|---|---|---|---|---|---|
| MobileNetV2 | $d$=−4.59, $p_{\mathrm{Bonf}}$=0.0025 | $d$=−0.85, $p_{\mathrm{Bonf}}$=0.659 | $d$=−0.29, $p_{\mathrm{Bonf}}$=1.0 (equiv.) | $d$=+0.61, $p_{\mathrm{Bonf}}$=1.0 | $d$=+1.10, $p_{\mathrm{Bonf}}$=0.350 (equiv.) |
| EfficientNet-B0 | $d$=−2.54, $p_{\mathrm{Bonf}}$=0.0237 | $d$=−0.28, $p_{\mathrm{Bonf}}$=1.0 | $d$=−3.17, $p_{\mathrm{Bonf}}$=0.0104 | $d$=−0.71, $p_{\mathrm{Bonf}}$=0.951 (equiv.) | $d$=−0.17, $p_{\mathrm{Bonf}}$=1.0 (equiv.) |
| GhostNet-100 | $d$=−1.33, $p_{\mathrm{Bonf}}$=0.206 | $d$=−0.63, $p_{\mathrm{Bonf}}$=1.0 | $d$=−1.33, $p_{\mathrm{Bonf}}$=0.203 (equiv.) | $d$=+0.20, $p_{\mathrm{Bonf}}$=1.0 (equiv.) | $d$=+0.36, $p_{\mathrm{Bonf}}$=1.0 (equiv.) |
| ResNet-18 | $d$=−1.65, $p_{\mathrm{Bonf}}$=0.106 | $d$=−0.63, $p_{\mathrm{Bonf}}$=1.0 | $d$=−0.45, $p_{\mathrm{Bonf}}$=1.0 | $d$=−0.08, $p_{\mathrm{Bonf}}$=1.0 | $d$=−0.19, $p_{\mathrm{Bonf}}$=1.0 |
| ShuffleNetV2-1.0x | $d$=−2.68, $p_{\mathrm{Bonf}}$=0.0196 | $d$=−0.33, $p_{\mathrm{Bonf}}$=1.0 | $d$=+0.03, $p_{\mathrm{Bonf}}$=1.0 | $d$=−0.26, $p_{\mathrm{Bonf}}$=1.0 | $d$=−0.17, $p_{\mathrm{Bonf}}$=1.0 |

# Appendix F Rank-5, AUC, and EER for the main comparison

Section 3.5 notes that rank-5 accuracy, AUC, and EER are computed and retained alongside rank-1 accuracy for every evaluation in this paper but not presented in the main text, which reports rank-1 identity accuracy throughout to keep the paper's length manageable. Table 30 reports these three additional metrics for Table 4's central comparison (`lr`, `span_baseline`, `span_tiny`; EarVN1.0, $n = 5$ seeds, all eleven backbones), as mean (rank-5) or mean±std (AUC, EER) across seeds. These are reported descriptively, at the same $n = 5$ seeds as Table 4, rather than re-analyzed with this paper's full paired-test/Bonferroni/TOST battery; the pattern across all three metrics is consistent with Table 4's rank-1 finding throughout (small, inconsistently-signed `span_tiny`-vs-`span_baseline` differences relative to the `lr`-vs-SR gap), and no case shows a qualitatively different picture from rank-1 accuracy for any backbone.

**Table 30** Rank-5 accuracy, AUC, and EER for Table 4's central comparison, EarVN1.0, $n = 5$ seeds, all eleven recognition backbones. Rank-5 is a mean across seeds (no per-seed std recorded); AUC and EER are mean±std across seeds. lr = no-SR, base. = span_baseline, tiny = span_tiny.

| Backbone | Rank-5 | | | AUC | | | EER | | |
|---|---|---|---|---|---|---|---|---|---|
| | lr | base. | tiny | lr | base. | tiny | lr | base. | tiny |
| EfficientNet-B0 | 0.7597 | 0.7891 | 0.7820 | 0.8602±0.0031 | 0.8759±0.0032 | 0.8754±0.0023 | 0.2213±0.0030 | 0.2053±0.0026 | 0.2059±0.0024 |
| GhostNet-100 | 0.7096 | 0.7499 | 0.7439 | 0.8419±0.0018 | 0.8670±0.0035 | 0.8642±0.0015 | 0.2365±0.0017 | 0.2138±0.0037 | 0.2161±0.0016 |
| LCNet-100 | 0.6811 | 0.7312 | 0.7171 | 0.8141±0.0069 | 0.8428±0.0012 | 0.8317±0.0050 | 0.2633±0.0064 | 0.2377±0.0006 | 0.2478±0.0051 |
| MobileNetV2 | 0.7024 | 0.7510 | 0.7464 | 0.8550±0.0071 | 0.8752±0.0053 | 0.8754±0.0026 | 0.2240±0.0060 | 0.2037±0.0049 | 0.2035±0.0026 |
| MobileNetV3-Large | 0.7040 | 0.7497 | 0.7435 | 0.8289±0.0055 | 0.8500±0.0067 | 0.8449±0.0029 | 0.2505±0.0048 | 0.2312±0.0064 | 0.2362±0.0023 |
| MobileNetV3-Small | 0.6547 | 0.6868 | 0.6827 | 0.7980±0.0014 | 0.8136±0.0031 | 0.8134±0.0040 | 0.2770±0.0008 | 0.2643±0.0028 | 0.2648±0.0038 |
| MobileOne-S0 | 0.7190 | 0.7425 | 0.7447 | 0.8405±0.0065 | 0.8542±0.0057 | 0.8535±0.0033 | 0.2385±0.0051 | 0.2264±0.0053 | 0.2268±0.0036 |
| RegNetY-400MF | 0.7081 | 0.7513 | 0.7502 | 0.8439±0.0021 | 0.8686±0.0015 | 0.8677±0.0027 | 0.2359±0.0022 | 0.2127±0.0017 | 0.2132±0.0021 |
| ResNet-18 | 0.6979 | 0.7162 | 0.7159 | 0.8544±0.0037 | 0.8648±0.0061 | 0.8646±0.0034 | 0.2267±0.0038 | 0.2171±0.0055 | 0.2168±0.0044 |
| ShuffleNetV2-1.0x | 0.6937 | 0.7274 | 0.7276 | 0.8276±0.0034 | 0.8496±0.0032 | 0.8475±0.0027 | 0.2511±0.0028 | 0.2313±0.0037 | 0.2324±0.0027 |
| SqueezeNet-1.1 | 0.6694 | 0.6980 | 0.6957 | 0.8072±0.0036 | 0.8189±0.0061 | 0.8163±0.0067 | 0.2691±0.0030 | 0.2590±0.0059 | 0.2609±0.0058 |

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